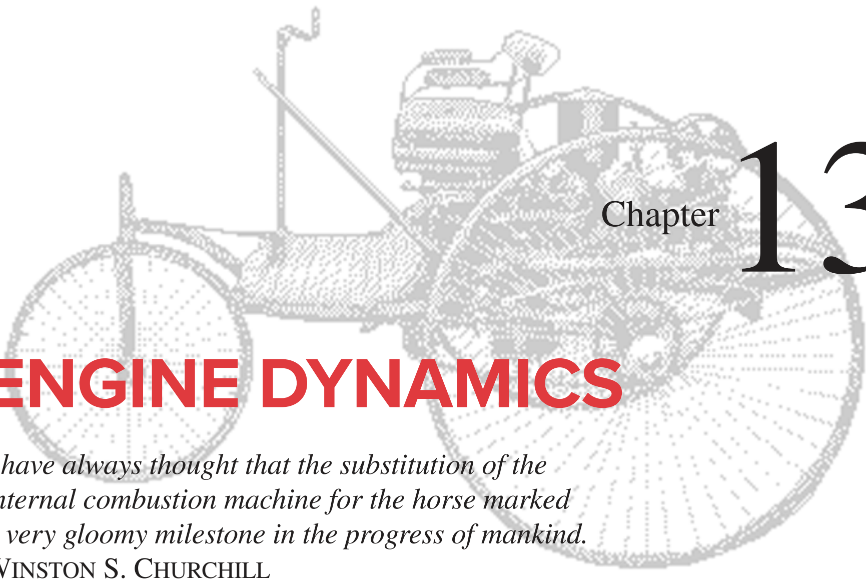
Chapter 13



ENGINE DYNAMICS

I have always thought that the substitution of the internal combustion machine for the horse marked a very gloomy milestone in the progress of mankind.
WINSTON S. CHURCHILL

13.0 INTRODUCTION *Watch a Video on Engine Kinematics* (48:17)[*]

The previous chapters have introduced analysis techniques for the determination of dynamic forces, moments, and torques in machinery. Shaking forces, moments, and their balancing have also been discussed. We will now attempt to integrate all these dynamic considerations into the design of a common device, the slider-crank linkage as used in the internal combustion engine.[†] This deceptively simple mechanism will be found to be actually quite complex in terms of the dynamic considerations necessary to its design for high-speed operation. Thus it will serve as an excellent example of the application of the dynamics concepts just presented. We will not address the thermodynamic aspects of the internal combustion engine beyond defining the combustion forces which are necessary to drive the device. Many other texts, such as those listed in the bibliography at the end of this chapter, deal with the very complex thermodynamic and fluid dynamic aspects of this ubiquitous device. We will concentrate only on its kinematics and mechanical dynamics aspects. It is not our intention to make an "engine designer" of the student so much as to apply dynamic principles to a realistic design problem of general interest and also to convey the complexity and fascination involved in the design of an apparently simple dynamic device.

Some students may have had the opportunity to disassemble and service an internal combustion engine, but many have never done so. Thus we will begin with very fundamental descriptions of engine design and operation. The program LINKAGES, supplied with this text, is designed to reinforce and amplify the concepts presented. It will perform all the tedious computations necessary to provide the student with dynamic information

[*] http://www.designofmachinery.com/DOM/Engine_Kinematics.mp4

[†] Basic information on the operation of engines with animations can be found at: *http://www.Howstuffworks.com/engine.htm*

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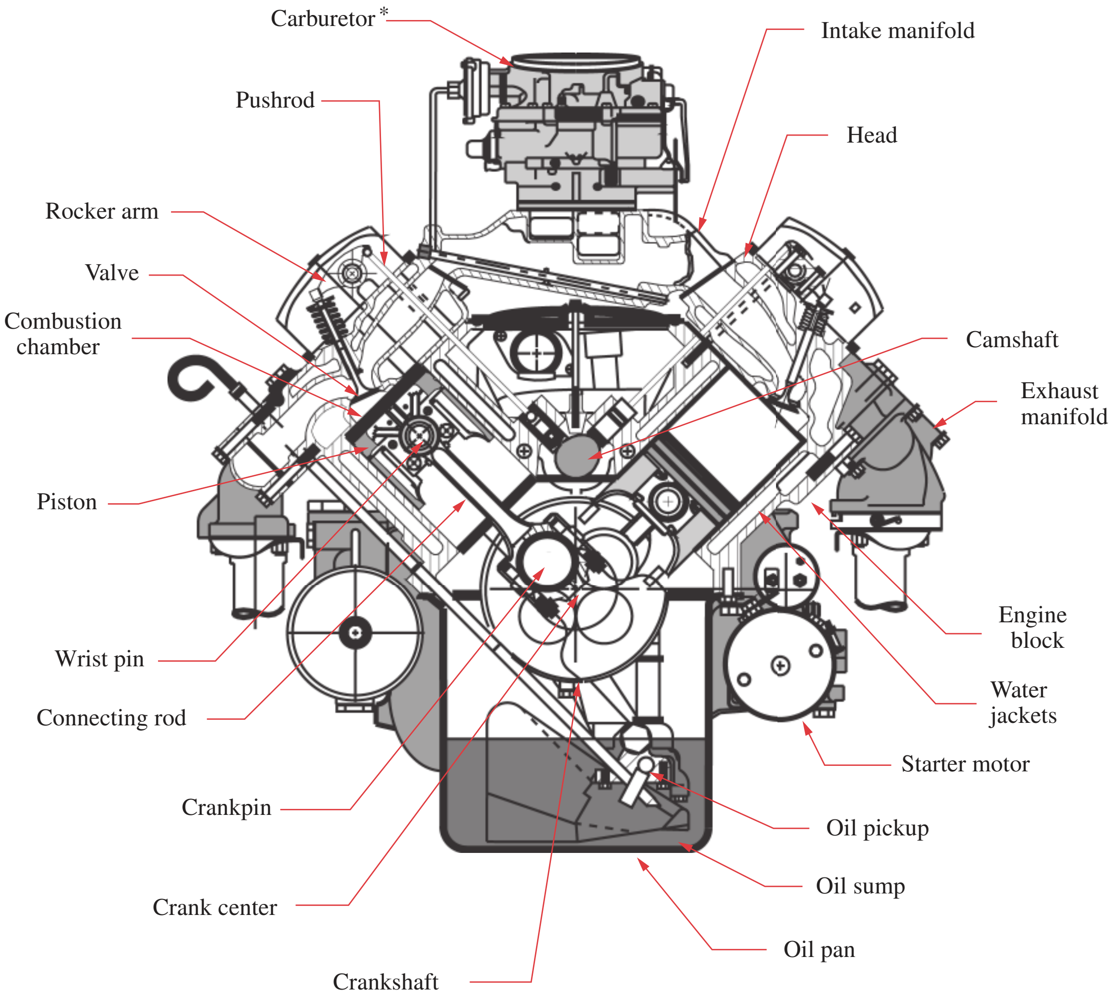

FIGURE 13-1

Cutaway cross section of a vee-eight engine

Source: Adapted from a drawing by Lane Thomas, Western Carolina University, Dept. of Industrial Education, with permission

for design choices and trade-offs. The student is encouraged to use this program concurrent with a reading of the text. Many examples and illustrations within the text will be generated with this program and reference will frequently be made to it. A user manual for program LINKAGES is within the program. Use it to gain familiarity with the program's operation. Examples used in Chapters 13 and 14 that deal with engine dynamics are built into the ENGINE routine of program LINKAGES for student observation and exercise. They can be found on a dropdown menu in that program. Other example engine files for program LINKAGES are downloadable.

* Carburetors have been replaced by fuel injection systems on automotive and other engines that are required to meet increasingly stringent exhaust emission control regulations in the United States and other countries. Fuel injection gives better control over the fuel-air mixture than a carburetor.

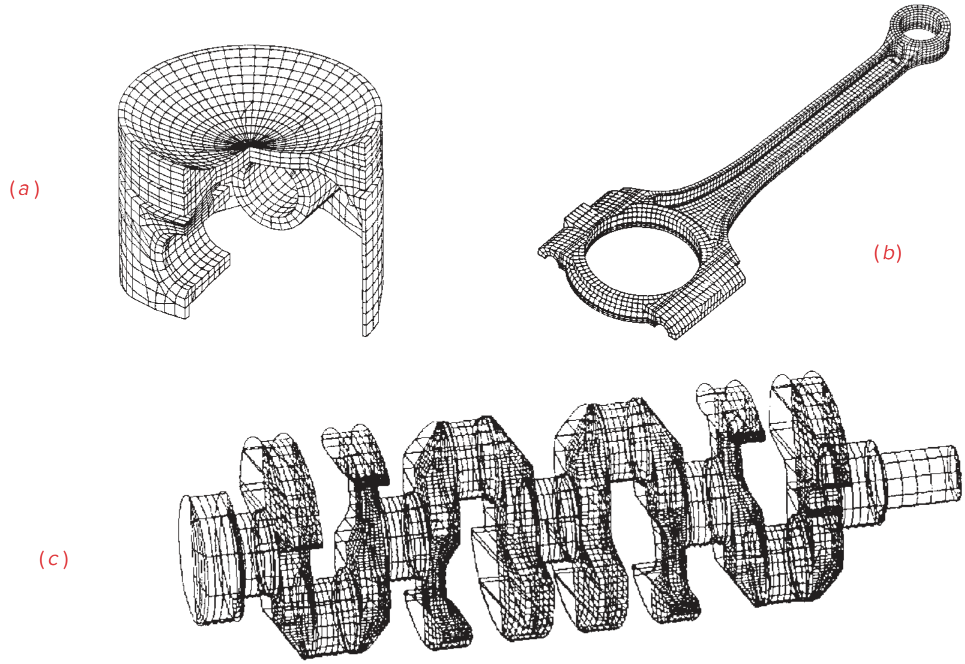

FIGURE 13-2

Finite-element models of an engine piston (*a*), connecting rod (*b*), and crankshaft (*c*) *Source: General Motors Co.*

13.1 ENGINE DESIGN

Figure 13-1 shows a detailed cross section of an internal combustion engine. The basic mechanism consists of a crank, a connecting rod (coupler), and piston (slider). Since this figure depicts a **multicylinder vee-eight** engine configuration, there are four cranks arranged on a crankshaft, and eight sets of connecting rods and pistons, four in the left bank of cylinders and four in the right bank. Only two piston-connecting rod assemblies are visible in this view, both on a common crank pin. The others are behind those shown. Figure 13-2 shows finite-element models of a piston, connecting rod, and crankshaft for a four-cylinder inline engine. The most usual arrangement is an inline engine with cylinders all in a common plane. Three-, four-, five-, and six-cylinder **inline engines** are in production the world over. **Vee engines** in four-, six-, eight-, ten-, and twelve-cylinder versions are also in production, with vee six and vee eight being the most popular vee configurations. The geometric arrangements of the crankshaft and cylinders have a significant effect on the dynamic condition of the engine. We will explore these effects of multicylinder arrangements in the next chapter. At this stage we wish to deal only with the design of a **single-cylinder** engine. After optimizing the geometry and dynamic condition of one cylinder, we will be ready to assemble combinations of cylinders into multicylinder configurations.

A schematic of the basic **one-cylinder slider-crank** mechanism and the terminology for its principal parts are shown in Figure 13-3. Note that it is "backdriven" compared to

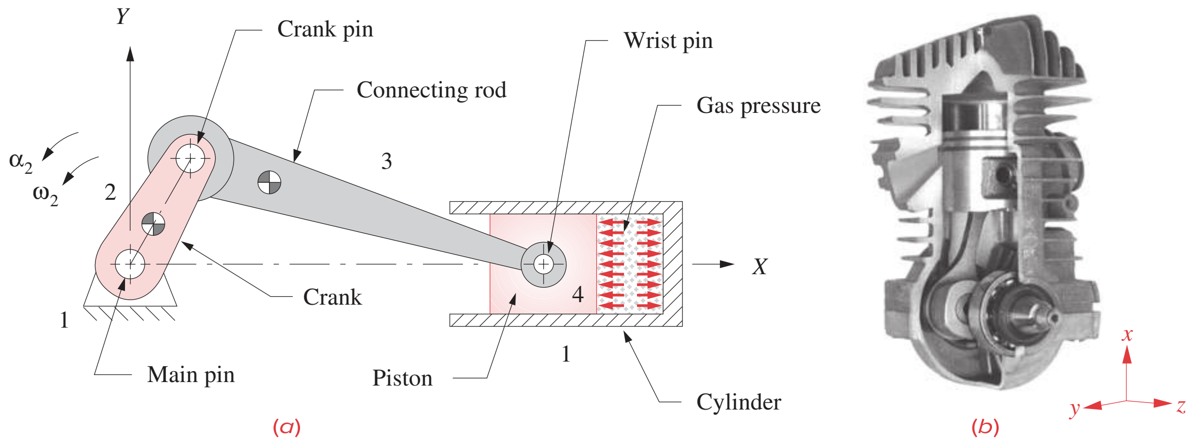

FIGURE 13-3

Fourbar slider-crank mechanism (*a*) for single-cylinder internal combustion engine (*b*) *Mahle Inc., Morristown, NJ*

the linkages we have been analyzing in previous chapters. That is, the explosion of the combustible mixture in the cylinder drives the piston to the left in Figure 13-3 or down in Figure 13-4, turning the crank. The crank torque that results is ultimately delivered to the drive wheels of the vehicle through a transmission (see Section 9.11), to propel the car, motorcycle, or other device. The same slider-crank mechanism can also be used "forward-driven," by motor-driving the crank and taking the output energy from the piston end. It is then called a **piston pump** and is used to compress air and pump well water, gasoline, and other liquids.

In the internal combustion engine of Figure 13-3, it should be fairly obvious that at most we can only expect energy to be delivered from the exploding gases to the crank during the power stroke of the cycle. The piston must return from bottom dead center (BDC) to top dead center (TDC) on its own momentum before it can receive another push from the next explosion. In fact, some rotational kinetic energy must be stored in the crankshaft merely to carry it through the TDC and BDC points as the moment arm for the gas force at those points is zero. This is why an internal combustion engine must be "spun up" with a hand crank, pull rope, or starter motor to get it running.

There are two common combustion cycles in use in internal combustion engines, the **Clerk two-stroke cycle** and the **Otto four-stroke cycle**, named after their nineteenth century inventors. The four-stroke cycle is most common in automobile, truck, and stationary gasoline engines. The two-stroke cycle is used in motorcycles, outboard motors, chain saws, and other applications where its better power-to-weight ratio outweighs its drawbacks of higher pollution levels and poor fuel economy compared to the four-stroke.

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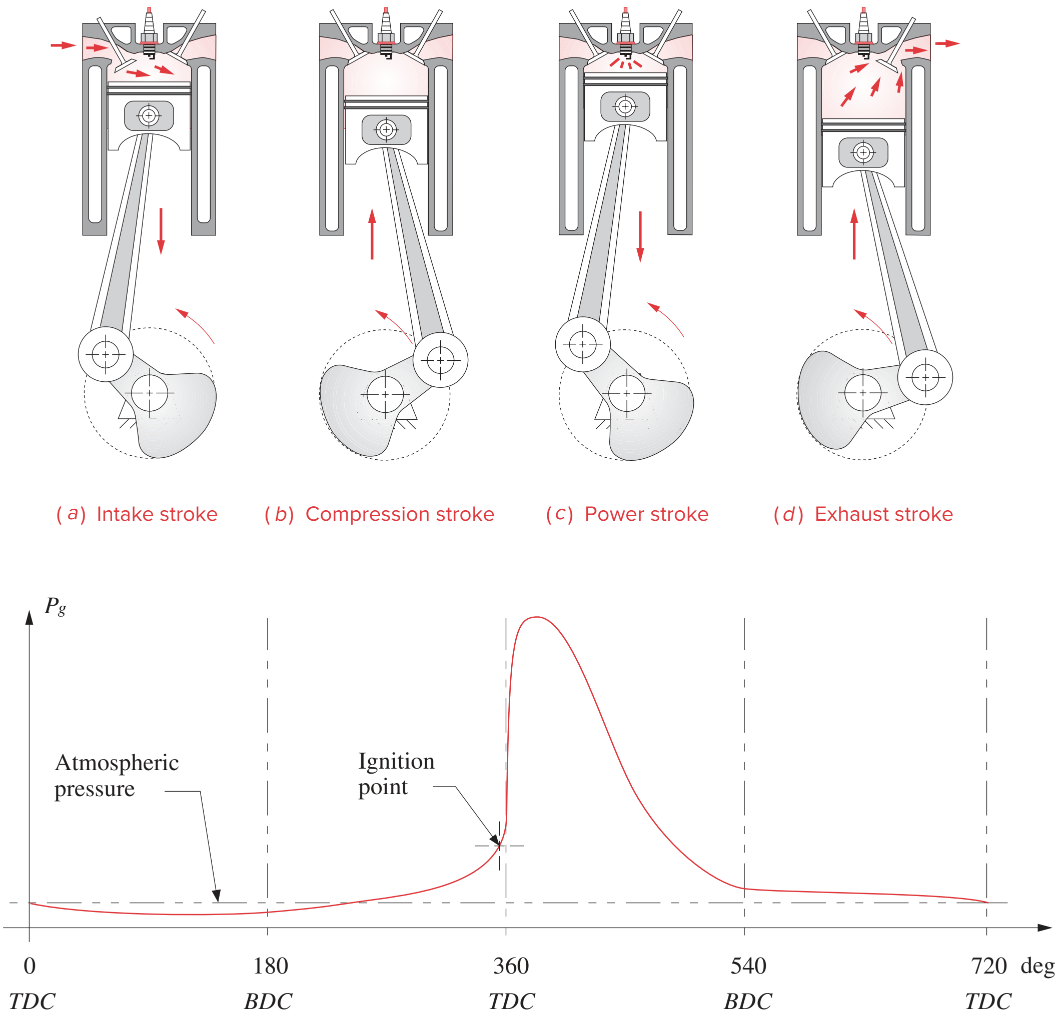
FOUR-STROKE CYCLE The **Otto four-stroke cycle** is shown in Figure 13-4. It takes four full strokes of the piston to complete one Otto cycle. A piston stroke is defined as its travel from TDC to BDC or the reverse. Thus there are two strokes per 360° crank revolution, and it takes 720° of crankshaft rotation to complete one four-stroke cycle. This engine requires at least two valves per cylinder, one for intake and one for exhaust. For discussion, we can start the cycle at any point as it repeats every two crank revolutions.



(*a*) Intake stroke (*b*) Compression stroke (*c*) Power stroke (*d*) Exhaust stroke

(*e*) The gas pressure curve

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FIGURE 13-4

The Otto four-stroke combustion cycle

Figure 13-4a shows the **intake stroke** which starts with the piston at TDC. A mixture of fuel and air is drawn into the cylinder from the induction system (the fuel injectors, or the carburetor and intake manifold in Figure 13-1), as the piston descends to BDC, increasing the volume of the cylinder and creating a slight negative pressure.

During the **compression stroke** in Figure 13-4b, all valves are closed and the gas is compressed as the piston travels from BDC to TDC. Slightly before TDC, a spark is ignited to explode the compressed gas. The pressure from this explosion builds very quickly and pushes the piston down from TDC to BDC during the **power stroke** shown in Figure 13-4c. The exhaust valve is opened and the piston's **exhaust stroke** from BDC to TDC (Figure 13-4d) pushes the spent gases out of the cylinder into the exhaust manifold (see also Figure 13-1) and thence to the catalytic converter for cleaning before being dumped out the tailpipe. The cycle is then ready to repeat with another intake stroke. The valves are opened and closed at the right times in the cycle by a camshaft which is driven in synchrony with the crankshaft by gears, chain, or toothed belt drive. (See Figure 9-25.) Figure 13-4e shows the gas pressure curve for one cycle. With a one-cylinder Otto cycle engine, power is delivered to the crankshaft, at most, 25% of the time as there is only 1 power stroke per 2 revolutions.

TWO-STROKE CYCLE The **Clerk two-stroke cycle** is shown in Figure 13-5. This engine does not need any valves, though to increase its efficiency, it is sometimes provided with a passive (pressure differential operated) one at the intake port. It does not have a camshaft or valve train or cam drive gears to add weight and bulk to the engine. As its name implies, it requires only two strokes, or 360°, to complete its cycle. There is a passageway, called a transfer port, between the combustion chamber above the piston and the crankcase below. There is also an exhaust port in the side of the cylinder. The piston acts to sequentially block or expose these ports as it moves up and down. The crankcase is sealed and mounts the carburetor on it, serving also as the intake manifold.

Starting at TDC (Figure 13-5a), the two-stroke cycle proceeds as follows: The spark plug ignites the fuel-air charge, compressed in the previous revolution. The expansion of the burning gases drives the piston down, delivering torque to the crankshaft. Partway down, the piston uncovers the exhaust port, allowing the burned (and also some unburned) gases to begin to escape to the exhaust system.

As the piston descends (Figure 13-5b), it compresses the charge of fuel-air mixture in the sealed crankcase. The piston blocks the intake port, preventing blowback through the carburetor. As the piston clears the transfer port in the cylinder wall, its downward motion pushes the new fuel-air charge up through the transfer port to the combustion chamber. The momentum of the exhaust gases leaving the chamber on the other side helps pull in the new charge as well.

The piston passes BDC (Figure 13-5c) and starts up, pushing out the remaining exhaust gases. The exhaust port is closed by the piston as it ascends, allowing compression of the new charge. As the piston approaches TDC, it exposes the intake port (Figure 13-5d), sucking a new charge of air and fuel into the expanded crankcase from the carburetor. Slightly before TDC, the spark is ignited and the cycle repeats as the piston passes TDC.

Clearly, this Clerk cycle is not as efficient as the Otto cycle in which each event is more cleanly separated from the others. Here there is much mixing of the various phases of the cycle. Unburned hydrocarbons are exhausted in larger quantities. This accounts for the poor fuel economy and dirty emissions of the Clerk engine.* It is nevertheless popular in applications where low weight is paramount.

Lubrication is also more difficult in the two-stroke engine than in the four-stroke as the crankcase is not available as an oil sump. Thus the lubricating oil must be mixed with the fuel. This further increases the emissions problem compared to the Otto cycle engine which burns raw gasoline and pumps its lubricating oil separately throughout the engine.

* Research and development is underway to clean up the emissions of the two-stroke engine by using fuel injection and compressed air scavenging of the cylinders. These efforts may yet bring this potentially more powerful engine design into compliance with air quality specifications.

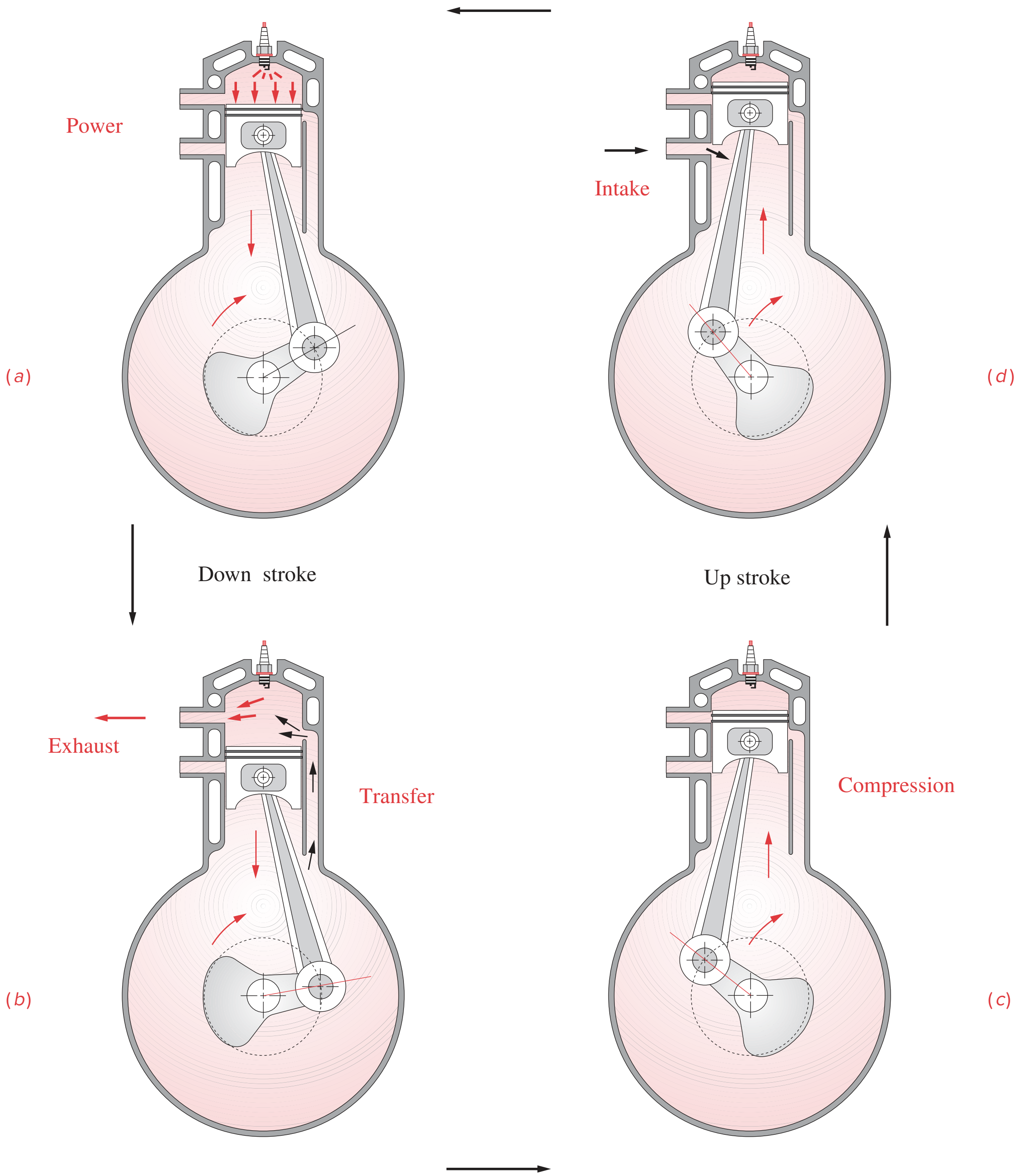

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FIGURE 13-5

The Clerk two-stroke combustion cycle

Diesel Cycle The **diesel cycle** can be either two-stroke or four-stroke. It is a **compression-ignition** cycle. No spark is needed to ignite the air-fuel mixture. The air is compressed in the cylinder by a factor of about 14 to 15 or more (versus 8 to 10 in the spark engine), and a low volatility fuel is injected into the cylinder just before TDC. The heat of compression causes the explosion. Diesel engines are larger and heavier than spark ignition engines for the same power output because the higher pressures and forces at which they operate require stronger, heavier parts. Two-stroke cycle diesel engines are quite common. Diesel fuel is a better lubricant than gasoline.

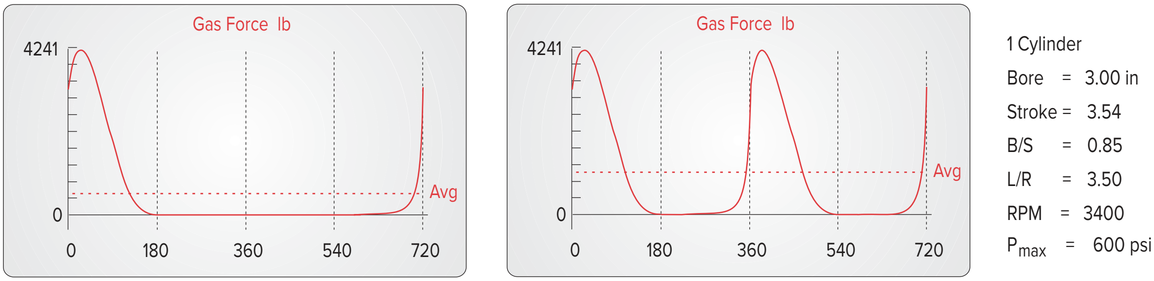
Gas Force In all the engines discussed here, the usable **output torque** is created from the explosive gas pressure generated within the cylinder either once or twice per two revolutions of the crank, depending on the cycle used. The magnitude and shape of this explosion pressure curve will vary with the engine design, stroke cycle, fuel used, speed of operation, and other factors related to the thermodynamics of the system. For our purpose of analyzing the mechanical dynamics of the system, we need to keep the gas pressure function consistent while we vary other design parameters in order to compare the results of our mechanical design changes. For this purpose, program LINKAGES has been provided with a built-in **gas pressure curve** whose peak value is about 600 psi and whose shape is similar to the curve from a real engine. Figure 13-6 shows the **gas force curve** that results from the built-in gas pressure function in program LINKAGES applied to a piston of particular area, for both two- and four-stroke engines. Changes in piston area will obviously affect the gas force magnitude for this consistent pressure function, but no changes in engine design parameters input to this program will change its built-in gas pressure curve. To see this gas force curve, run program LINKAGES and select any one of the example engines from the pulldown menu. Then calculate and plot *Gas Force*.

13.2 SLIDER-CRANK KINEMATICS

In Chapters 4, 6, 7, and 11 we developed general equations for the exact solution of the positions, velocities, accelerations, and forces in the pin-jointed fourbar linkage, and also for two inversions of the **slider-crank linkage**, using vector equations. We could again apply that method to the analysis of the "standard" slider-crank linkage, used in the majority of internal combustion engines, as shown in Figure 13-7. Note that its slider motion has been aligned with the X axis. This is a "nonoffset" slider-crank, because the slider axis extended passes through the crank pivot. It also has its slider block translating against the stationary ground plane; thus there will be no Coriolis component of acceleration (see Section 7.3).

The simple geometry of this particular inversion of the slider-crank mechanism allows a very straightforward approach to the exact analysis of its slider's position, velocity, and acceleration, using only plane trigonometry and scalar equations. Because of this method's simplicity and to present an alternative solution approach we will analyze this device again.

Let the crank radius be r and the conrod length be l. The angle of the crank is θ, and the angle that the conrod makes with the X axis is ϕ. For any constant crank angular velocity ω, the crank angle $\theta = \omega t$. The instantaneous piston position is x. Two right triangles rqs and lqu are constructed. Then from geometry:



(*a*) Otto four-stroke cycle (*b*) Clerk two-stroke cycle

FIGURE 13-6

Gas force functions in the two-stroke and four-stroke cycle engines

$$q = r\sin\theta = l\sin\phi$$
$$\theta = \omega t \tag{13.1a}$$
$$\sin\phi = \frac{r}{l}\sin\omega t$$

$$s = r\cos\omega t$$
$$u = l\cos\phi \tag{13.1b}$$
$$x = s + u = r\cos\omega t + l\cos\phi$$

$$\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1-\left(\frac{r}{l}\sin\omega t\right)^2} \tag{13.1c}$$

$$x = r\cos\omega t + l\sqrt{1-\left(\frac{r}{l}\sin\omega t\right)^2} \tag{13.1d}$$

Equation 13.1d is an exact expression for the piston position x as a function of r, l, and ωt. This can be differentiated versus time to obtain exact expressions for the velocity and acceleration of the piston. For a steady-state analysis we will assume ω to be constant.

$$\dot{x} = -r\omega\left[\sin\omega t + \frac{r}{2l}\frac{\sin 2\omega t}{\sqrt{1-\left(\frac{r}{l}\sin\omega t\right)^2}}\right] \tag{13.1e}$$

$$\ddot{x} = -r\omega^2\left\{\cos\omega t - \frac{r\left[l^2\left(1-2\cos^2\omega t\right)-r^2\sin^4\omega t\right]}{\left[l^2-\left(r\sin\omega t\right)^2\right]^{\frac{3}{2}}}\right\} \tag{13.1f}$$

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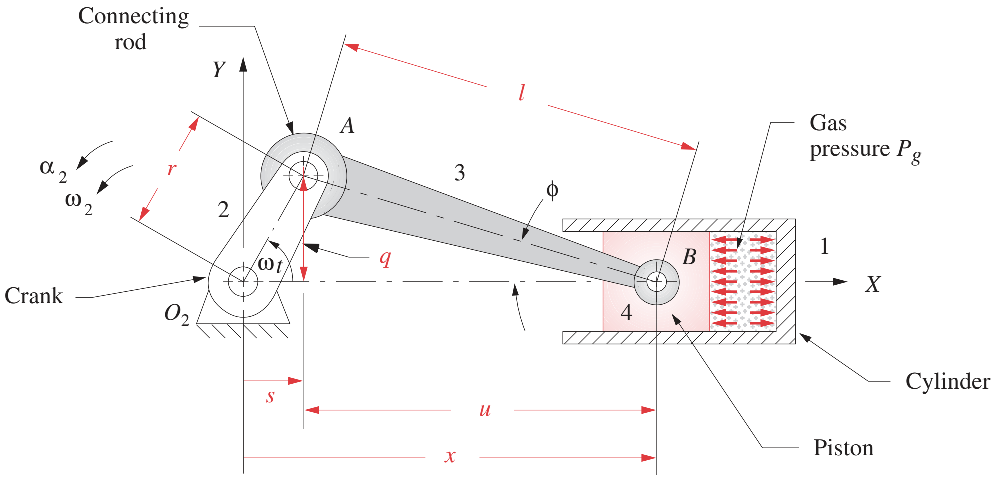
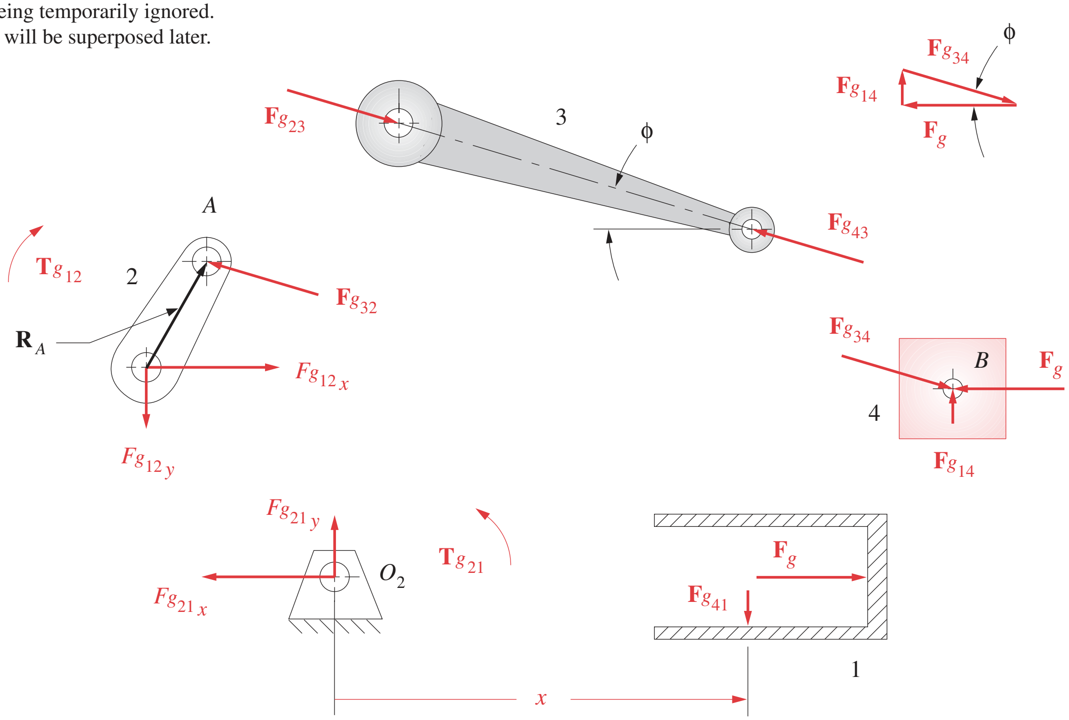


Note:
Link 3 can be considered as a 2-force member for this gas-force analysis because the inertia forces are being temporarily ignored. They will be superposed later.

(*a*) The linkage geometry



(*b*) Free-body diagrams

FIGURE 13-7

Fourbar slider-crank linkage position and gas force analysis (See Figure 13-12 for the inertia force analysis.)

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Equations 13.1 can easily be solved with a computer for all values of ωt needed. But, it is rather difficult to look at equation 13.1f and visualize the effects of changes in the design parameters r and l on the acceleration. It would be useful if we could derive a simpler expression, even if approximate, that would allow us to more easily predict the results of design decisions involving these variables. To do so, we will use the binomial theorem to expand the radical in equation 13.1d for piston position to put the equations for position, velocity, and acceleration in simpler, approximate forms which will shed some light on the dynamic behavior of the mechanism.

The general form of the binomial theorem is:

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \cdots \tag{13.2a}$$

The radical in equation 13.1d is:

$$\sqrt{1-\left(\frac{r}{l}\sin\omega t\right)^2} = \left[1-\left(\frac{r}{l}\sin\omega t\right)^2\right]^{\frac{1}{2}} \tag{13.2b}$$

where, for the binomial expansion:

$$a = 1 \qquad b = -\left(\frac{r}{l}\sin\omega t\right)^2 \qquad n = \frac{1}{2} \tag{13.2c}$$

It expands to:

$$1-\frac{1}{2}\left(\frac{r}{l}\sin\omega t\right)^2 - \frac{1}{8}\left(\frac{r}{l}\sin\omega t\right)^4 - \frac{1}{16}\left(\frac{r}{l}\sin\omega t\right)^6 - \cdots \tag{13.2d}$$

or:

$$1-\left(\frac{r^2}{2l^2}\right)\sin^2\omega t - \left(\frac{r^4}{8l^4}\right)\sin^4\omega t - \left(\frac{r^6}{16l^6}\right)\sin^6\omega t - \cdots \tag{13.2e}$$

Each nonconstant term contains the **crank-conrod ratio** r/l to some power. Applying some engineering common sense to the depiction of the slider-crank in Figure 13-7a, we can see that if r/l were greater than 1 the crank could not make a complete revolution. In fact if r/l even gets close to 1, the piston will hit the fixed pivot O_2 before the crank completes its revolution. If r/l is as large as 1/2, the transmission angle $(\pi/2 - \phi)$ will be too small (see Sections 3.3 and 4.11) and the linkage will not run well. A practical upper limit on the value of r/l is about 1/3. Most slider-crank linkages will have this **crank-conrod ratio** somewhere between 1/3 and 1/5 for smooth operation. If we substitute this practicl upper limit of $r/l = 1/3$ into equation 13.2e, we get:

$$1-\left(\frac{1}{18}\right)\sin^2\omega t - \left(\frac{1}{648}\right)\sin^4\omega t - \left(\frac{1}{11\,664}\right)\sin^6\omega t - \cdots$$

(13.2f)

or:

$$1-0.055\,56\sin^2\omega t - 0.001\,54\sin^4\omega t - 0.000\,09\sin^6\omega t - \cdots$$

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Clearly we can drop all terms after the second with very small error. Substituting this approximate expression for the radical in equation 13.1d gives an approximate expression for piston displacement with only a fraction of one percent error.

$$x \cong r\cos\omega t + l\left[1-\left(\frac{r^2}{2l^2}\right)\sin^2\omega t\right] \tag{13.3a}$$

Substitute the trigonometric identity:

$$\sin^2\omega t = \frac{1-\cos 2\omega t}{2} \tag{13.3b}$$

and simplify:

$$x \cong l - \frac{r^2}{4l} + r\left(\cos\omega t + \frac{r}{4l}\cos 2\omega t\right) \tag{13.3c}$$

Differentiate for velocity of the piston (with constant ω):

$$\dot{x} \cong -r\omega\left(\sin\omega t + \frac{r}{2l}\sin 2\omega t\right) \tag{13.3d}$$

Differentiate again for acceleration (with constant ω):

$$\ddot{x} \cong -r\omega^2\left(\cos\omega t + \frac{r}{l}\cos 2\omega t\right) \tag{13.3e}$$

The process of binomial expansion has, in this particular case, led us to Fourier series approximations of the exact expressions for the piston displacement, velocity, and acceleration. Fourier* showed that any periodic function can be approximated by a series of sine and cosine terms of integer multiples of the independent variable. Recall that we dropped the fourth, sixth, and subsequent power terms from the binomial expansion, which would have provided cos 4ω*t*, cos 6ω*t*, etc., terms in this expression. These multiple-angle functions are referred to as the **harmonics** of the fundamental cos ω*t* term. The cos ω*t* term repeats once per crank revolution and is called the fundamental frequency or the **primary component**. The second harmonic, cos 2ω*t*, repeats twice per crank revolution and is called the **secondary component**. The higher harmonics were dropped when we truncated the series. The constant term in the displacement function is the **DC component** or **average value**. The complete function is the sum of its harmonics. The Fourier series form of the expressions for displacement and its derivatives lets us see the relative contributions of the various harmonic components of the functions. This approach will prove to be quite valuable when we attempt to dynamically balance an engine design.

Program LINKAGES calculates the position, velocity, and acceleration of the piston according to equations 13.3c, d, and e. Figure 13-8a, b, and c shows these functions for this example engine in the program as plotted for constant crank ω over two full revolutions. The acceleration curve shows the effects of the second harmonic term most clearly because that term's coefficient is larger than its correspondent in either of the other two functions. The fundamental (–cos ω*t*) term gives a pure harmonic function with a period of 360°. This fundamental term dominates the function as it has the largest coefficient in equation 13.3e. The flat top and slight dip in the positive peak acceleration of Figure

* Baron Jean Baptiste Joseph Fourier (1768-1830) published the description of the mathematical series which bears his name in *The Analytic Theory of Heat* in 1822. The Fourier series is widely used in harmonic analysis of all types of physical systems. Its general form is:

$$y = \frac{a_0}{2} + (a_1\cos x + b_1\sin x) + (a_2\cos 2x + b_2\sin 2x) + \cdots + (a_n\cos nx + b_n\sin nx)$$

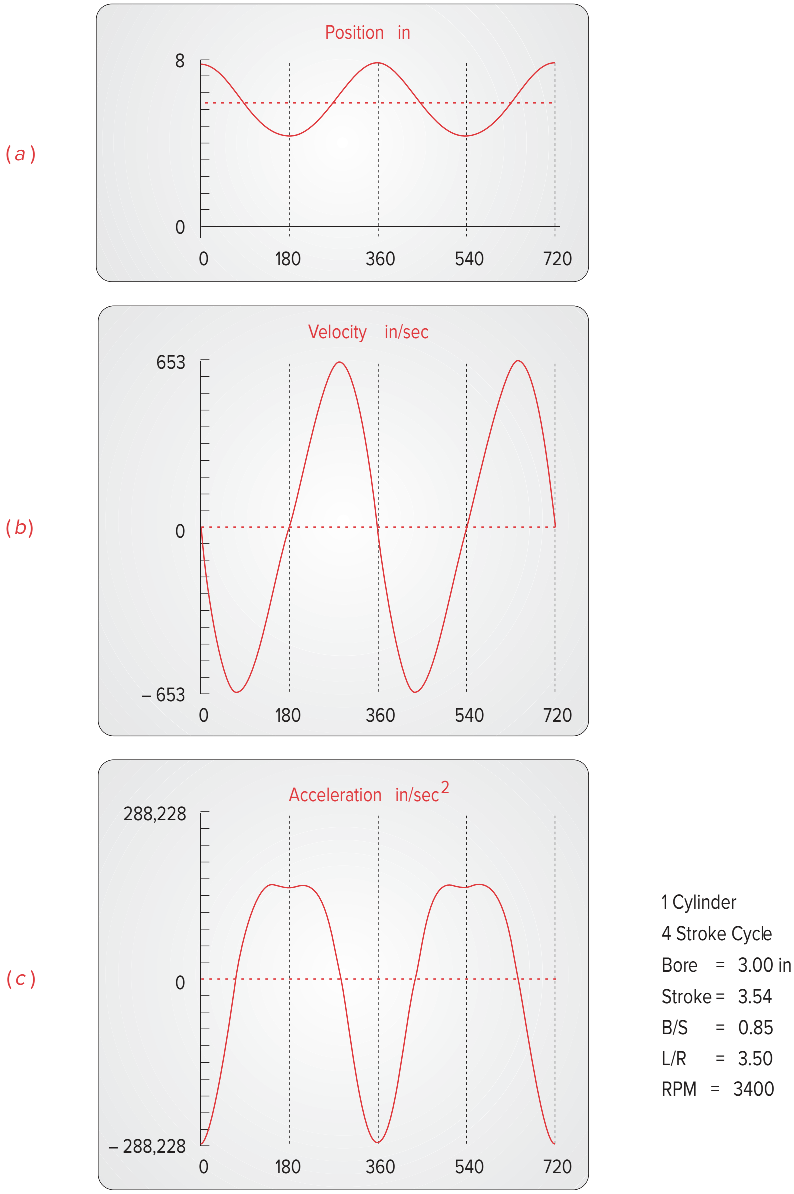


FIGURE 13-8

Position, velocity, and acceleration functions for a single-cylinder engine

13-8c are caused by the cos $2\omega t$ second harmonic adding or subtracting from the fundamental. Note the very high value of peak acceleration of the piston even at the midrange engine speed of 3400 rpm. It is 747 *g*'s! At 6000 rpm this increases to nearly 1300 *g*'s.* This is a moderately sized engine, of 3-in (76 mm) bore and 3.54-in (89 mm) stroke, with 25-in^3 (400-cc) displacement per cylinder (a 1.6-L 4-cylinder engine).

* If you think that number is high, consider the typical Nascar pushrod V-8 that turns up to 9600 rpm and Formula 1, V-12, and V-8 racing engines that red-line at over 19 000 rpm. As an exercise, calculate their peak accelerations assuming the same dimensions as our example engine.

SUPERPOSITION We will now analyze the dynamic behavior of the single-cylinder engine based on the approximate kinematic model developed in this section. Since we have several sources of dynamic excitation to deal with, we will use the method of superposition to separately analyze them and then combine their effects. We will first consider the **forces and torques** which are due to the presence of the **explosive gas forces** in the cylinder, which drive the engine. Then we will analyze the **inertia forces and torques** that result from the high-speed motion of the elements. The total force and torque state of the machine at any instant will be the sum of these components. Finally we will look at the **shaking forces and torques** on the ground plane and the **pin forces** within the linkage that result from the combination of applied and dynamic forces on the system.

13.3 GAS FORCE AND GAS TORQUE

The **gas force** is due to the gas pressure from the exploding fuel-air mixture impinging on the top of the piston surface as shown in Figure 13-3. Let F_g = gas force, P_g = gas pressure, A_p = area of piston, and B = bore of cylinder, which is also equal to the piston diameter. Then:

$$\mathbf{F}_g = -P_g A_p \, \hat{\mathbf{i}}; \qquad A_p = \frac{\pi}{4} B^2$$
$$\mathbf{F}_g = -\frac{\pi}{4} P_g B^2 \, \hat{\mathbf{i}} \tag{13.4}$$

The negative sign is due to the choice of engine orientation in the coordinate system of Figure 13-3. The **gas pressure** P_g in this expression is a function of crank angle ωt and is defined by the thermodynamics of the engine. A typical **gas pressure curve** for a four-stroke engine is shown in Figure 13-4. The **gas force curve** shape is identical to that of the gas pressure curve as they differ only by a constant multiplier, the piston area A_p. Figure 13-6 shows the approximation of the gas force curve used in program LINKAGES for both four- and two-stroke engines.

The **gas torque** in Figure 13-9 is due to the gas force acting at a moment arm about the crank center O_2 in Figure 13-7. This moment arm varies from zero to a maximum as the crank rotates. The distributed gas force over the piston surface has been resolved to a single force acting through the mass center of link 4 in the free-body diagrams of Figure 13-7b. The concurrent force system at point B is resolved in the vector diagram showing that:

$$\mathbf{F}_{g_{14}} = F_g \tan\phi \, \hat{\mathbf{j}} \tag{13.5a}$$

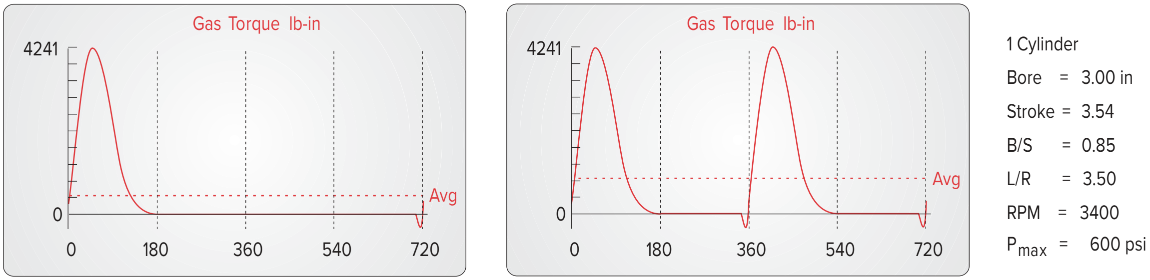
$$\mathbf{F}_{g_{34}} = -F_g \, \hat{\mathbf{i}} - F_g \tan\phi \, \hat{\mathbf{j}} \tag{13.5b}$$

From the free-body diagrams in Figure 13-7 we can see that:

$$\mathbf{F}_{g_{41}} = -\mathbf{F}_{g_{14}}$$
$$\mathbf{F}_{g_{43}} = -\mathbf{F}_{g_{34}}$$
$$\mathbf{F}_{g_{23}} = -\mathbf{F}_{g_{43}}$$
$$\mathbf{F}_{g_{32}} = -\mathbf{F}_{g_{23}}$$

so:

$$\mathbf{F}_{g_{32}} = -\mathbf{F}_{g_{34}} = F_g \, \hat{\mathbf{i}} + F_g \tan\phi \, \hat{\mathbf{j}}$$



(*a*) Otto four-stroke cycle

(*b*) Clerk two-stroke cycle

FIGURE 13-9

Gas torque functions in the two-stroke and four-stroke cycle engines

The **driving torque** $\mathbf{T}_{g_{21}}$ at link 2 due to the gas force can be found from the cross product of the position vector to point A and the force at point A.

$$\mathbf{T}_{g_{21}} = \mathbf{R}_A \times \mathbf{F}_{g_{32}} \tag{13.6a}$$

This expression can be expanded and will involve the crank length r and the angles θ and ϕ as well as the gas force $\mathbf{F}_g$. Note from the free-body diagram for link 1 that we can also express the torque in terms of the forces $\mathbf{F}_{g14}$ or $\mathbf{F}_{g41}$, which act always perpendicular to the motion of the slider (neglecting friction), and the distance x, which is their instantaneous moment arm about O_2. The reaction torque $\mathbf{T}_{g12}$ due to the gas force trying to rock the ground plane is:

$$\mathbf{T}_{g_{12}} = F_{g_{41}} \cdot x \ \hat{\mathbf{k}} \tag{13.6b}$$

If you have ever abruptly opened the throttle of a running automobile engine while working on it, you probably noticed the engine move to the side as it rocked in its mounts from the reaction torque. The driving torque $\mathbf{T}_{g21}$ is the negative of this reaction torque.

$$\mathbf{T}_{g_{21}} = -\mathbf{T}_{g_{12}}$$

$$\mathbf{T}_{g_{21}} = -F_{g_{41}} \cdot x \ \hat{\mathbf{k}} \tag{13.6c}$$

and: $$F_{g_{14}} = -F_{g_{41}}$$

so: $$\mathbf{T}_{g_{21}} = F_{g_{14}} \cdot x \ \hat{\mathbf{k}} \tag{13.6d}$$

Equation 13.6d gives us an expression for **gas torque** which involves the displacement of the piston x for which we have already derived equation 13.3a. Substituting equation 13.3a for x and the magnitude of equation 13.5a for F_{g14}, we get:

$$\mathbf{T}_{g_{21}} = \left(F_g \tan\phi\right)\left[l - \frac{r^2}{4l} + r\left(\cos\omega t + \frac{r}{4l}\cos 2\omega t\right)\right]\hat{\mathbf{k}} \tag{13.6e}$$

Equation 13.6e contains the conrod angle ϕ as well as the independent variable, crank angle ωt. We would like to have an expression which involves only ωt. We can substitute an expression for $\tan\phi$ generated from the geometry of Figure 13-7a.

$$\tan\phi = \frac{q}{u} = \frac{r\sin\omega t}{l\cos\phi} \tag{13.7a}$$

Substitute equation 13.1c for cos ϕ:

$$\tan\phi = \frac{r\sin\omega t}{l\sqrt{1-\left(\frac{r}{l}\sin\omega t\right)^2}} \tag{13.7b}$$

The radical in the denominator can be expanded using the binomial theorem as was done in equations 13.2, and the first two terms retained for a good approximation to the exact expression,

$$\frac{1}{\sqrt{1-\left(\frac{r}{l}\sin\omega t\right)^2}} \cong 1+\frac{r^2}{2l^2}\sin^2\omega t \tag{13.7c}$$

giving:

$$\tan\phi \cong \frac{r}{l}\sin\omega t\left(1+\frac{r^2}{2l^2}\sin^2\omega t\right) \tag{13.7d}$$

Substitute this into equation 13.6e for the gas torque:

$$\mathbf{T}_{g_{21}} \cong F_g\left[\frac{r}{l}\sin\omega t\left(1+\frac{r^2}{2l^2}\sin^2\omega t\right)\right]\left[l-\frac{r^2}{4l}+r\left(\cos\omega t+\frac{r}{4l}\cos 2\omega t\right)\right]\hat{\mathbf{k}} \tag{13.8a}$$

Expand this expression and neglect any terms containing the conrod crank ratio r/l raised to any power greater than one since these will have very small coefficients as was seen in equation 13.2f. This results in a simpler, but even more approximate expression for the gas torque:

$$\mathbf{T}_{g_{21}} \cong F_g\, r\sin\omega t\left(1+\frac{r}{l}\cos\omega t\right) \tag{13.8b}$$

Note that the **exact value** of this **gas torque** can always be calculated from equations 13.1d, 13.5a, and 13.6d in combination, or from the expansion of equation 13.6a, if you require a more accurate answer. For design purposes the approximate equation 13.8b will usually be adequate. Program LINKAGES calculates the gas torque using equation 13.8b and its built-in gas pressure curve to generate the gas force function. Plots of the gas torque for two- and four-stroke cycles are shown in Figure 13-9. Note the similarity in shape to that of the gas force curve in Figure 13-6. Note also that the two-stroke has theoretically twice the power available as the four-stroke, all other factors being equal, because there are twice as many torque pulses per unit time. The poorer efficiency of the two-stroke significantly reduces this theoretical advantage, however.

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13.4 EQUIVALENT MASSES *Watch a Video on Engine Dynamics (53:17)**

To do a complete dynamic force analysis on any mechanism, we need to know the geometric properties (mass, center of gravity, mass moment of inertia) of the moving links

* http://www.designofmachinery.com/DOM/Engine_Dynamics.mp4

is easy to do if the link already is designed in detail and its dimensions are known. When designing the mechanism from scratch, we typically do not yet know that level of detail about the links' geometries. But we must nevertheless make some estimate of their geometric parameters in order to begin the iteration process which will eventually converge on a detailed design.

In the case of this slider-crank mechanism, the **crank** is in **pure rotation** and the **piston** is in **pure translation**. By assuming some reasonable geometries and materials, we can make approximations of their dynamic parameters. Their kinematic motions are easily determined. We have already derived expressions for the piston motion in equations 13.3. Further, if we balance the rotating crank, as described and recommended in the previous chapter, then the *CG* of the crank will be motionless at its center O_2 and will not contribute to the dynamic forces. We will do this in a later section.

The conrod is in complex motion. To do an exact dynamic analysis as was derived in Section 11.5, we need to determine the linear acceleration of its *CG* for all positions. At the outset of the design, the conrod's *CG* location is not accurately defined. To "bootstrap" the design, we need a simplified model of this connecting rod which we can later refine, as more dynamic information is generated about our engine design. The requirements for a dynamically equivalent model were stated in Section 10.2 and are repeated here as Table 13-1 for your convenience.

If we could model our still-to-be-designed conrod as two, lumped, point masses, concentrated one at the crank pin (point *A* in Figure 13-7), and one at the wrist pin (point *B* in Figure 13-7), we would at least know what the motions of these lumps are. The lump at *A* would be in pure rotation as part of the crank, and the lump at point *B* would be in pure translation as part of the piston. These lumped, point masses have no dimension and are assumed to be connected with a magical, massless but rigid rod.*

* These lumped mass models have to be made with *very* special materials. *Unobtainium 206* has the property of infinite mass density, thus occupies no space and can be used for "point masses." *Unobtainium 208* has infinite stiffness and zero mass and thus can be used for rigid but "massless rods."

DYNAMICALLY EQUIVALENT MODEL Figure 13-10a shows a typical conrod. Figure 13-10b shows a generic two-mass model of the conrod. One mass m_t is located at distance l_t from the *CG* of the original rod, and the second mass m_p at distance l_p from the *CG*. The mass of the original part is m_3, and its moment of inertia about its *CG* is I_{G_3}. Expressing the three requirements for dynamic equivalence from Table 13-1 mathematically in terms of these variables, we get:

$$m_p + m_t = m_3 \tag{13.9a}$$

$$m_p l_p = m_t l_t \tag{13.9b}$$

$$m_p l_p^2 + m_t l_t^2 = I_{G_3} \tag{13.9c}$$

There are four unknowns in these three equations, m_p, l_p, m_t, l_t, which means we must choose a value for any one variable to solve the system. Let us choose the distance

TABLE 13-1 Requirements for Dynamic Equivalence

1	The mass of the model must equal that of the original body.
2	The center of gravity must be in the same location as that of the original body.
3	The mass moment of inertia must equal that of the original body.

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l_t equal to the distance to the wrist pin, l_b, as shown in Figure 13-10c. This will put one mass at a desired location. Solving equations 13.9a and 13.9b simultaneously with that substitution gives expressions for the two lumped masses:

$$m_p = m_3 \frac{l_b}{l_p + l_b}$$
$$m_b = m_3 \frac{l_p}{l_p + l_b} \tag{13.9d}$$

Substituting equation 13.9d into 13.9c gives a relation between l_p and l_b:

$$m_3 \frac{l_b}{l_p + l_b} l_p^2 + m_3 \frac{l_p}{l_p + l_b} l_b^2 = I_{G_3} = m_3 l_p l_b$$
$$l_p = \frac{I_{G_3}}{m_3 l_b} \tag{13.9e}$$

Please refer to Section 10.10 and equation 10.13 that define the *center of percussion* and its geometric relationship to a corresponding *center of rotation*. Equation 13.9e is the same as equation 11.13 (except for sign which is due to an arbitrary choice of the link's orientation in the coordinate system). The distance l_p is the location of the center of percussion corresponding to a center of rotation at l_b. Thus our second mass m_p must be placed at the link's **center of percussion** P (using point B as its center of rotation) to obtain exact dynamic equivalence. The masses must be as defined in equation 13.9d.

The geometry of the typical conrod, as shown in Figures 13-2 and 13-10a, is large at the crank pin end (A) and small at the wrist pin end (B). This puts the CG close to the "big end." The center of percussion P will be even closer to the big end than is the CG. For this reason, we can place the second lumped mass, which belongs at P, at point A with relatively small error in our dynamic model's accuracy. This approximate model is adequate for our initial design calculations. Once a viable design geometry is established, we will have to do a complete and exact force analysis with the methods of Chapter 11 before considering the design complete.

Making this substitution of distance l_a for l_p and renaming the lumped masses at those distances m_{3a} and m_{3b}, to reflect both their identity with link 3 and with points A and B, we rewrite equations 13.9d.

Let $$l_p = l_a$$

then: $$m_{3a} = m_3 \frac{l_b}{l_a + l_b} \tag{13.10a}$$

and: $$m_{3b} = m_3 \frac{l_a}{l_a + l_b} \tag{13.10b}$$

These define the amounts of the total conrod mass to be placed at each end, to approximately dynamically model that link. Figure 13-10d shows this dynamic model. In the absence of any data on the shape of the conrod at the outset of a design, preliminary dynamic force information can be obtained by using the rule of thumb of placing two-thirds of the conrod's mass at the crank pin and one-third at the wrist pin.

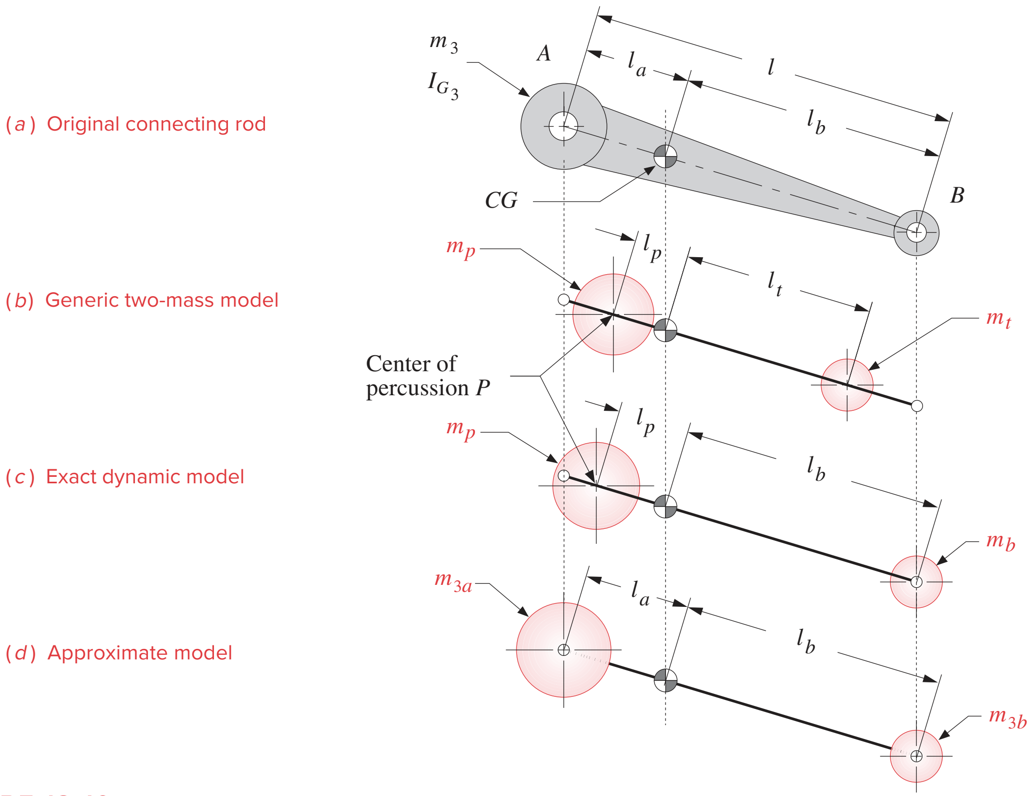

FIGURE 13-10

Lumped mass dynamic models of a connecting rod

STATICALLY EQUIVALENT MODEL We can create a similar lumped mass model of the crank. Even though we intend to balance the crank before we are done, for generality we will initially model it *unbalanced* as shown in Figure 13-11. Its *CG* is located at some distance r_{G_2} from the pivot, O_2, on the line to the crank pin, *A*. We would like to model it as a lumped mass at *A* on a massless rod pivoted at O_2. If our principal concern is with a steady-state analysis, then the crank velocity ω will be held constant. An absence of angular acceleration on the crank allows a statically equivalent model to be used because the equation $\mathbf{T} = I\alpha$ will be zero regardless of the value of *I*. A **statically equivalent model** needs only to have equivalent mass and equivalent first moments as shown in Table 13-2. The moments of inertia need not match. We model it as two lumped masses, one at point *A* and one at the fixed pivot O_2. Writing the two requirements for static equivalence from Table 13-2:

$$
\begin{aligned}
m_2 &= m_{2a} + m_{2O_2} \\
m_{2a} r &= m_2 r_{G_2} \\
m_{2a} &= m_2 \frac{r_{G_2}}{r}
\end{aligned}
\tag{13.11}
$$

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TABLE 13-2 Requirements for Static Equivalence

1	The mass of the model must equal that of the original body.
2	The center of gravity must be in the same location as that of the original body.

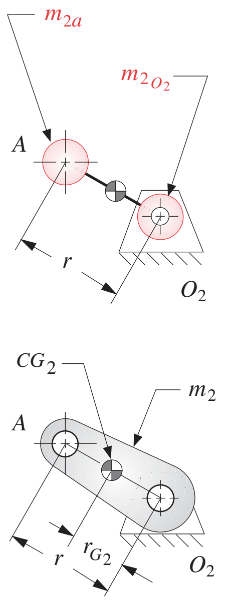


FIGURE 13-11
Statically equivalent lumped mass model of a crank

The lumped mass m_{2a} can be placed at point A to represent the unbalanced crank. The second lumped mass, at the fixed pivot O_2, is not necessary to any calculations as that point is stationary.

These simplifications lead to the lumped parameter model of the slider-crank linkage shown in Figure 13-12. The crank pin, point A, has two masses concentrated at it, the equivalent mass of the crank m_{2a} and the portion of conrod m_{3a}. Their sum is m_A. At the wrist pin, point B, two masses are also concentrated, the piston mass m_4 and the remaining portion of the conrod mass m_{3b}. Their sum is m_B. This model has masses which are either in pure rotation (m_A) or in pure translation (m_B), so it is very easy to dynamically analyze.

$$\begin{aligned} m_A &= m_{2a} + m_{3a} \\ m_B &= m_{3b} + m_4 \end{aligned} \tag{13.12}$$

Value of Models *The value of constructing simple, lumped mass models of complex systems increases with the complexity of the system being designed.* It makes little sense to spend large amounts of time doing sophisticated, detailed analyses of designs which are so ill-defined at the outset that their conceptual viability is as yet unproven. It is better to get a reasonably approximate and rapid answer that tells you the concept needs to be rethought than to spend a greater amount of time reaching the same conclusion to more decimal places.

13.5 INERTIA AND SHAKING FORCES

The simplified, lumped mass model of Figure 13-12 can be used to develop expressions for the forces and torques due to the accelerations of the masses in the system. The method of d'Alembert is of value in visualizing the effects of these moving masses on the system and on the ground plane. Accordingly, the free-body diagrams of Figure 13-12b show the d'Alembert inertia forces acting on the masses at points A and B. Friction is again ignored. The acceleration for point B is given in equation 13.3e. The acceleration of point A in pure rotation is obtained by differentiating the position vector $\mathbf{R}_A$ twice, assuming a constant crankshaft ω, which gives:

$$\begin{aligned} \mathbf{R}_A &= r\cos\omega t\,\hat{\mathbf{i}} + r\sin\omega t\,\hat{\mathbf{j}} \\ \mathbf{a}_A &= -r\omega^2\cos\omega t\,\hat{\mathbf{i}} - r\omega^2\sin\omega t\,\hat{\mathbf{j}} \end{aligned} \tag{13.13}$$

The total inertia force $\mathbf{F}_i$ is the sum of the centrifugal (inertia) force at point A and the inertia force at point B.

$$\mathbf{F}_i = -m_A\,\mathbf{a}_A - m_B\,\mathbf{a}_B \tag{13.14a}$$

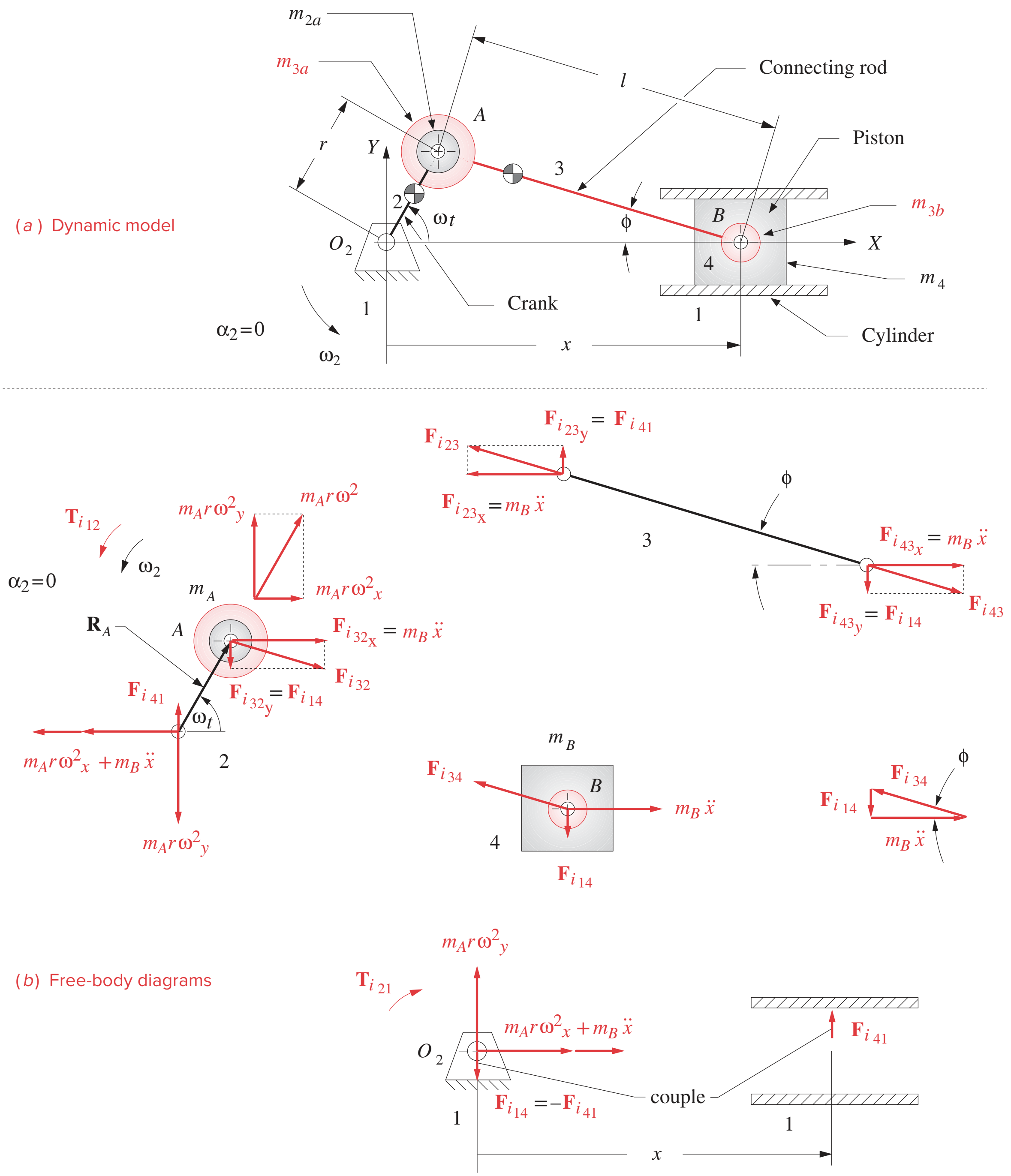


FIGURE 13-12

Lumped mass dynamic model of the slider-crank—arrows show vector direction and sense, labels show magnitude

Breaking it into x and y components:

$$F_{ix} = -m_A\left(-r\omega^2 \cos\omega t\right) - m_B\,\ddot{x} \tag{13.14b}$$

$$F_{iy} = -m_A\left(-r\omega^2 \sin\omega t\right) \tag{13.14c}$$

Note that only the x component is affected by the acceleration of the piston. Substituting equation 13.3e into equation 13.14b:

$$\begin{aligned} F_{ix} &\cong -m_A\left(-r\omega^2 \cos\omega t\right) - m_B\left[-r\omega^2\left(\cos\omega t + \frac{r}{l}\cos 2\omega t\right)\right] \\ F_{iy} &= -m_A\left(-r\omega^2 \sin\omega t\right) \end{aligned} \tag{13.14d}$$

Notice that the x-directed inertia forces have primary components at crank frequency and secondary (second harmonic) forces at twice crank frequency. There are also small-magnitude, higher, even harmonics which we truncated in the binomial expansion of the piston displacement function. The force due to the rotating mass at point A has only a primary component.

The **shaking force** was defined in Section 11.8 to be *the sum of all forces acting on the ground plane*. From the free-body diagram for link 1 in Figure 13-12:

$$\begin{aligned} \sum F_{sx} &\cong -m_A\left(r\omega^2 \cos\omega t\right) - m_B\left[r\omega^2\left(\cos\omega t + \frac{r}{l}\cos 2\omega t\right)\right] \\ \sum F_{sy} &= -m_A\left(r\omega^2 \sin\omega t\right) + F_{i_{41}} - F_{i_{41}} \end{aligned} \tag{13.14e}$$

Note that the side force of the piston on the cylinder wall $F_{i_{41}}$ is cancelled by an equal and opposite force $F_{i_{14}}$ passed through the connecting rod and crankshaft to the main pin at O_2. These two forces create a couple that provides the shaking torque. The shaking force $\mathbf{F}_s$ is equal to the negative of the inertia force.

$$\mathbf{F}_s = -\mathbf{F}_i \tag{13.14f}$$

Note that the gas force from equation 13.4 does not contribute to the shaking force. Only inertia forces and external forces are felt as shaking forces. The gas force is an internal force which is cancelled within the mechanism. It acts equally and oppositely on both the piston top and the cylinder head as shown in Figure 13-7.

Program LINKAGES calculates the shaking force at constant ω for any combination of linkage parameters input to it. Figure 13-13 shows the shaking force plot for the same unbalanced built-in example engine as shown in the acceleration plot (Figure 13-8c). The linkage orientation is the same as in Figure 13-12 with the x axis horizontal. The x component is larger than the y component due to the high acceleration of the piston. The forces are seen to be quite large despite this being a relatively small (0.4 liter per cylinder) engine running at moderate speed (3400 rpm). We will soon investigate techniques to reduce or eliminate this shaking force from the engine. It is an undesirable feature which creates noise and vibration.

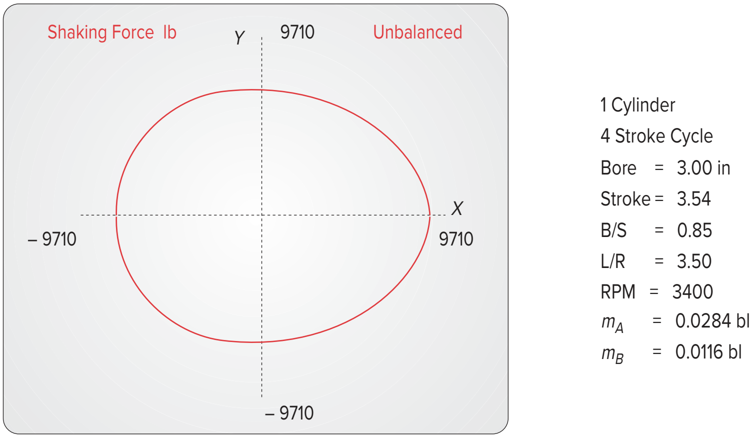


FIGURE 13-13

Shaking force in an unbalanced slider-crank linkage

13.6 INERTIA AND SHAKING TORQUES

The **inertia torque** results from the action of the inertia forces at a moment arm. The inertia force at point A in Figure 13-12 has two components, radial and tangential. The radial component has no moment arm. The tangential component has a moment arm of crank radius r. If the crank ω is constant, the mass at A will not contribute to inertia torque. The inertia force at B has a nonzero component perpendicular to the cylinder wall except when the piston is at TDC or BDC. As we did for the gas torque, we can express the inertia torque in terms of the couple $-\mathbf{F}_{i_{41}}, \mathbf{F}_{i_{41}}$ whose forces act always perpendicular to the motion of the slider (neglecting friction), and the distance x, which is their instantaneous moment arm (see Figure 13-12). The inertia torque is:

$$\mathbf{T}_{i_{21}} = \left(F_{i_{41}} \cdot x\right) \hat{\mathbf{k}} = \left(-F_{i_{14}} \cdot x\right) \hat{\mathbf{k}} \tag{13.15a}$$

Substituting for $F_{i_{14}}$ (see Figure 13-12b) and for x (see equation 13.3a), we get:

$$\mathbf{T}_{i_{21}} \cong -\left(-m_B\,\ddot{x}\tan\phi\right)\left[l - \frac{r^2}{4l} + r\left(\cos\omega t + \frac{r}{4l}\cos 2\omega t\right)\right]\hat{\mathbf{k}} \tag{13.15b}$$

We previously developed expressions for $\ddot{x}$ (equation 13.3e) and tan ϕ (equation 13.7d) which can now be substituted.

$$\begin{aligned}\mathbf{T}_{i_{21}} \cong m_B &\left[-r\omega^2\left(\cos\omega t + \frac{r}{l}\cos 2\omega t\right)\right] \\ \cdot &\left[\frac{r}{l}\sin\omega t\left(1 + \frac{r^2}{2l^2}\sin^2\omega t\right)\right] \\ \cdot &\left[l - \frac{r^2}{4l} + r\left(\cos\omega t + \frac{r}{4l}\cos 2\omega t\right)\right]\hat{\mathbf{k}}\end{aligned} \tag{13.15c}$$

Expanding this and then dropping all terms with coefficients containing *r/l* to powers higher than one gives the following approximate equation for inertia torque with constant shaft ω:

$$\mathbf{T}_{i_{21}} \cong -m_B r^2 \omega^2 \sin\omega t\left(\frac{r}{2l}+\cos\omega t+\frac{3r}{2l}\cos 2\omega t\right)\hat{\mathbf{k}} \tag{13.15d}$$

This contains products of sine and cosine terms. Putting it entirely in terms of harmonics will be instructive, so substitute the identities:

$$2\sin\omega t\cos 2\omega t = \sin 3\omega t - \sin\omega t$$
$$2\sin\omega t\cos\omega t = \sin 2\omega t$$

to get:

$$\mathbf{T}_{i_{21}} \cong \frac{1}{2} m_B r^2 \omega^2\left(\frac{r}{2l}\sin\omega t - \sin 2\omega t - \frac{3r}{2l}\sin 3\omega t\right)\hat{\mathbf{k}} \tag{13.15e}$$

This shows that the **inertia torque** has a third harmonic term as well as a first and second. The second harmonic is the dominant term as it has the largest coefficient because *r/l* is always less than 2/3.

The **shaking torque** is equal to the inertia torque.

$$\mathbf{T}_s = \mathbf{T}_{i_{21}} \tag{13.15f}$$

Program LINKAGES calculates the inertia torque from equation 13.15e. Figure 13-14 shows a plot of the inertia torque for this built-in example engine. Note the dominance of the second harmonic. The ideal magnitude for the inertia torque is zero, as it is parasitic. Its average value is always zero, so *it contributes nothing to the net driving torque*. It merely creates large positive and negative oscillations in the total torque which increase vibration and roughness. We will soon investigate means to reduce or eliminate this inertia and shaking torque in our engine designs. It is possible to cancel their effects by proper arrangement of the cylinders in a multicylinder engine, as will be explored in the next chapter.

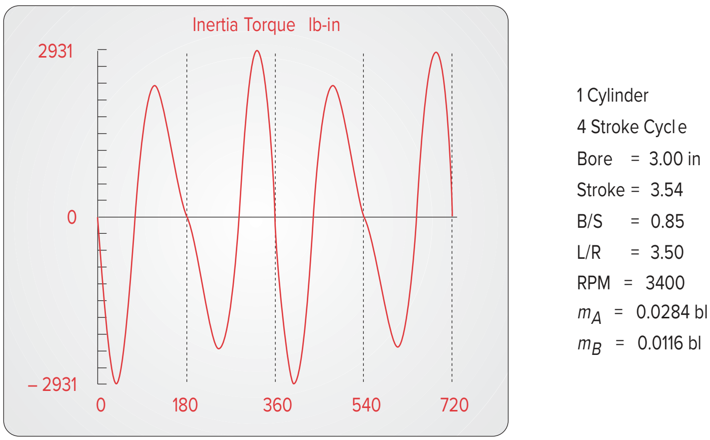


FIGURE 13-14
Inertia torque in the slider-crank linkage

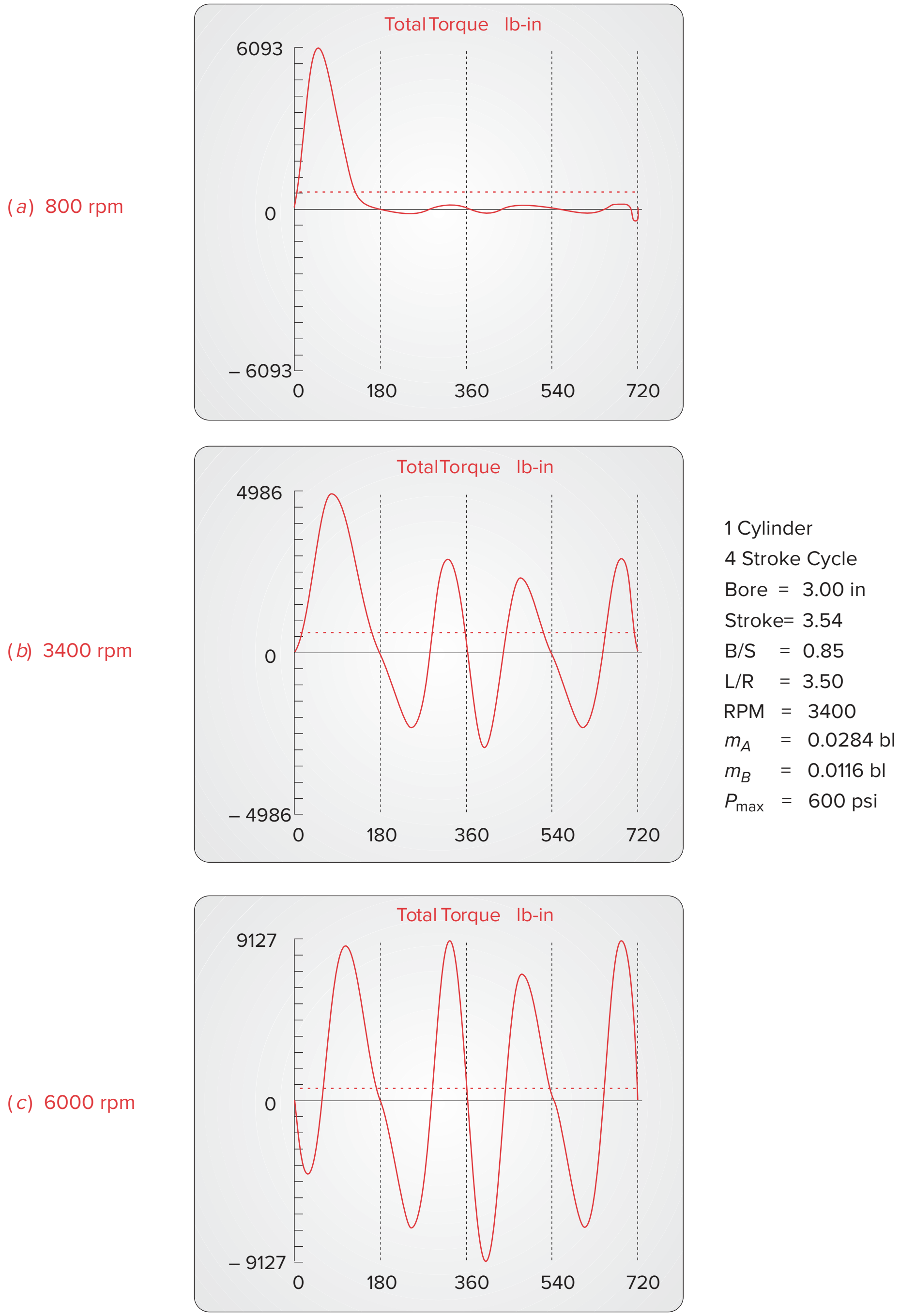


FIGURE 13-15

The total torque function's shape and magnitude vary with crankshaft speed

13.7 TOTAL ENGINE TORQUE

The total engine torque is the sum of the gas torque and the inertia torque.

$$\mathbf{T}_{total} = \mathbf{T}_g + \mathbf{T}_i \tag{13.16}$$

The gas torque is less sensitive to engine speed than is the inertia torque, which is a function of ω^2. So the relative contributions of these two components to the total torque will vary with engine speed. Figure 13-15a shows the total torque for this example engine plotted by program LINKAGES for an idle speed of 800 rpm. Compare this to the gas torque plot of the same engine in Figure 13-9a. The inertia torque component is negligible at this slow speed compared to the gas torque component. Figure 13-15c shows the same engine run at 6000 rpm. Compare this to the plot of inertia torque in Figure 13-14. The inertia torque component is dominating at this high speed. At the midrange speed of 3400 rpm (Figure 13-15b), a mix of both components is seen.

13.8 FLYWHEELS *Watch a Video on Balancing and Pin Forces (42:38)**

We saw in Section 11.11 that large oscillations in the torque-time function can be significantly reduced by the addition of a flywheel to the system. The single-cylinder engine is a prime candidate for the use of a flywheel. The intermittent nature of its power strokes makes one mandatory as it will store the kinetic energy needed to carry the piston through the Otto cycle's exhaust, intake, and compression strokes during which work must be done on the system. Even the two-stroke engine needs a flywheel to drive the piston up on the compression stroke.

The procedure for designing an engine flywheel is identical to that described in Section 11.11 for the fourbar linkage. The total torque function for one revolution of the crank is integrated, pulse by pulse, with respect to its average value. These integrals represent energy fluctuations in the system. The maximum change in energy under the torque curve during one cycle is the amount needed to be stored in the flywheel. Equation 11.20c expresses this relationship. Program LINKAGES does the numerical integration of the total torque function and presents a table similar to the one shown in Figure 11-11. These data and the designer's choice of a coefficient of fluctuation k (see equation 11.19b) are all that is needed to solve equations 11.20 and 11.21 for the required moment of inertia of the flywheel.

The calculation must be done at some average crank ω. Since the typical engine operates over a wide range of speeds, some thought needs to be given to the most appropriate speed to use in the flywheel calculation. The flywheel's stored kinetic energy is proportional to ω^2 (see equation 11.17). Thus at high speeds a flywheel can have a small moment of inertia and still be effective. The slowest operating speed will require the largest flywheel and should be the one used in the computation of required flywheel size.

Program LINKAGES plots the flywheel-smoothed total torque for a user-supplied coefficient of fluctuation k. Figure 13-16 shows the smoothed torque functions for $k = 0.05$ corresponding to the unsmoothed ones in Figure 13-15. Note that the smoothed curves shown for each engine speed are what would result with the flywheel size necessary to obtain that coefficient of fluctuation at that speed. In other words, the flywheel applied to the 800-rpm engine is much larger than the one on the 6000-rpm engine, in these plots.

* http://www.designofmachinery.com/DOM/Engine_Balancing_and_Pin_Forces.mp4

(a) 800 rpm

(b) 3400 rpm

(c) 6000 rpm

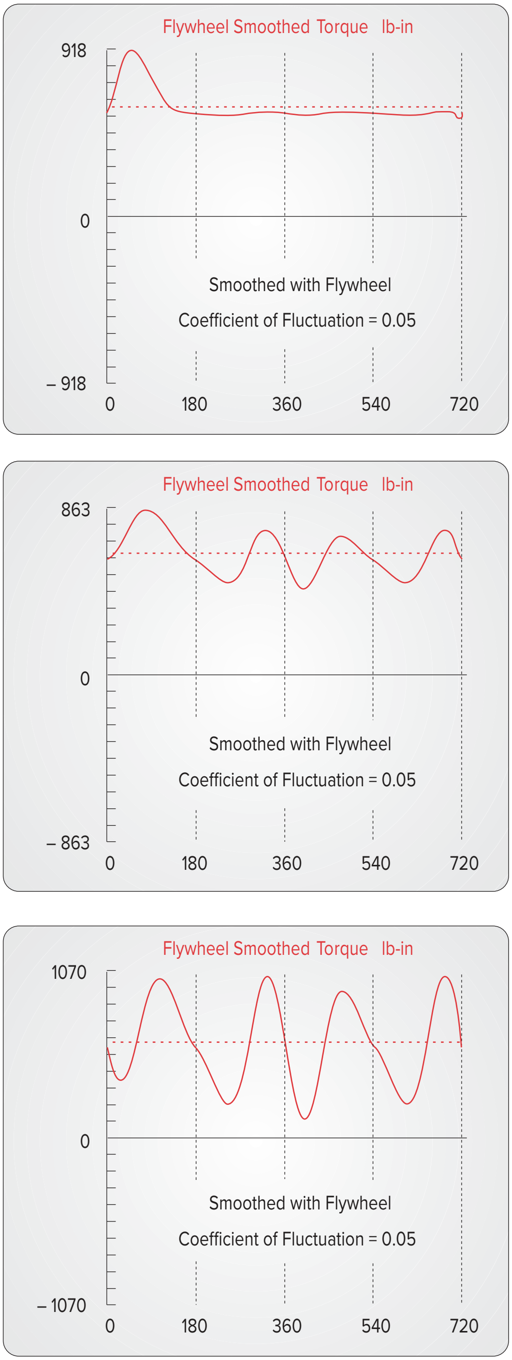


FIGURE 13-16

The total torque function's shape and magnitude vary with crankshaft speed

Compare corresponding rows (speeds) between Figures 13-15 and 13-16 to see the effect of the addition of a flywheel. But do not directly compare parts a, b, and c within Figure 13-16 as to the amount of smoothing since the flywheel sizes used are different at each operating speed.

An engine flywheel is usually designed as a flat disk, bolted to one end of the crankshaft. One flywheel face is typically used for the clutch to run against. The clutch is a friction device which allows disconnection of the engine from the drive train (the wheels of a vehicle) when no output is desired. The engine can then remain running at idle speed with the vehicle or output device stopped. When the clutch is engaged, all engine torque is transmitted through it, by friction, to the output shaft.

13.9 PIN FORCES IN THE SINGLE-CYLINDER ENGINE *Watch a Short Video on Pin Forces (20:03)**

* http://www.designofmachinery.com/DOM/Pin_Forces.mp4

In addition to calculating the overall effects on the ground plane of the dynamic forces present in the engine, we also need to know the magnitudes of the forces at the pin joints. These forces will dictate the design of the pins and the bearings at the joints. Though we were able to lump the mass due to both conrod and piston, or conrod and crank, at points *A* and *B* for an overall analysis of the linkage's effects on the ground plane, we cannot do so for the pin force calculations. This is because the pins feel the effect of the conrod pulling on one "side" and the piston (or crank) pulling on the other "side" of the pin as shown in Figure 13-17. Thus we must separate the effects of the masses of the links joined by the pins.

We will calculate the effect of each component due to the various masses and the gas force and then superpose them to obtain the complete pin force at each joint. We need a bookkeeping system to keep track of all these components. We have already used some subscripts for these forces, so we will retain them and add others. The resultant bearing loads have the following components:

1 The gas force components, with the subscript *g*, as in F_g.

2 The inertia force due to the piston mass, with subscript *ip*, as in F_{ip}.

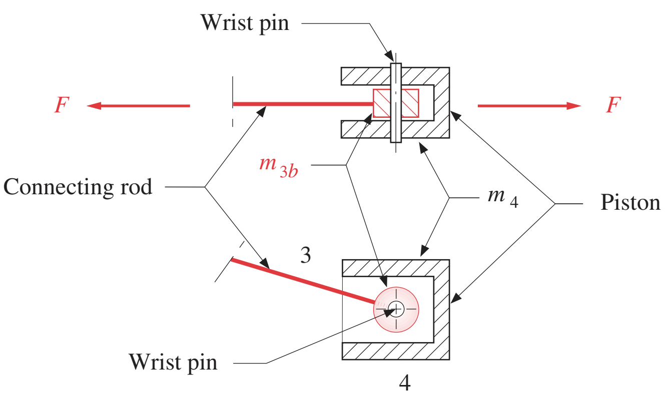


FIGURE 13-17

Forces on a pivot pin

3 The inertia force due to the mass of the conrod at the wrist pin, with subscript *iw*, as in F_{iw}.

4 The inertia force due to the mass of the conrod at the crank pin, with subscript *ic*, as in F_{ic}.

5 The inertia force due to the mass of the crank at the crank pin, with subscript *ir*, as in F_{ir}.

The link number designations will be added to each subscript in the same manner as before, indicating the link from which the force is coming as the first number and the link being analyzed as the second. (See Section 11.2 for further discussion of this notation.)

Figure 13-18 shows the free-body diagrams for the inertia force $\mathbf{F}_{ipB}$ due only to the acceleration of the mass of the piston, m_4. Those components are:

$$\mathbf{F}_{ip_B} = -m_4 a_B \hat{\mathbf{i}} \tag{13.17a}$$

$$\mathbf{F}_{ip_{14}} = -F_{ip_B} \tan\phi\, \hat{\mathbf{j}} = m_4 a_B \tan\phi\, \hat{\mathbf{j}} \tag{13.17b}$$

$$\mathbf{F}_{ip_{34}} = -\mathbf{F}_{ip_B} - \mathbf{F}_{ip_{14}} = m_4 a_B\, \hat{\mathbf{i}} - m_4 a_B \tan\phi\, \hat{\mathbf{j}} \tag{13.17c}$$

$$\mathbf{F}_{ip_{32}} = -\mathbf{F}_{ip_{34}} = -m_4 a_B\, \hat{\mathbf{i}} + m_4 a_B \tan\phi\, \hat{\mathbf{j}} \tag{13.17d}$$

$$\mathbf{F}_{ip_{12}} = -\mathbf{F}_{ip_{32}} = \mathbf{F}_{ip_{34}} \tag{13.17e}$$

Figure 13-19 shows the free-body diagrams for the forces due only to the acceleration of the mass of the conrod located at the wrist pin, m_{3b}. Those components are:

$$\mathbf{F}_{iw_B} = -m_{3b} a_B \hat{\mathbf{i}} \tag{13.18a}$$

$$\mathbf{F}_{iw_{34}} = \mathbf{F}_{iw_{41}} = F_{iw_B} \tan\phi\, \hat{\mathbf{j}} = -m_{3b} a_B \tan\phi\, \hat{\mathbf{j}} \tag{13.18b}$$

$$\mathbf{F}_{iw_{43}} = -\mathbf{F}_{iw_{34}} = m_{3b} a_B \tan\phi\, \hat{\mathbf{j}} \tag{13.18c}$$

$$\mathbf{F}_{iw_{23}} = -\mathbf{F}_{iw_B} - \mathbf{F}_{iw_{43}} = m_{3b} a_B\, \hat{\mathbf{i}} - m_{3b} a_B \tan\phi\, \hat{\mathbf{j}} \tag{13.18d}$$

$$\mathbf{F}_{iw_{12}} = -\mathbf{F}_{iw_{32}} = \mathbf{F}_{iw_{23}} \tag{13.18e}$$

Figure 13-20a shows the free-body diagrams for the forces due only to the acceleration of the mass of the conrod located at the crank pin, m_{3a}. That component is:

$$\mathbf{F}_{ic} = -\mathbf{F}_{ic_{12}} = \mathbf{F}_{ic_{21}} = -m_{3a} \mathbf{a}_A \tag{13.19a}$$

13

Substitute equation 13.13 for $\mathbf{a}_A$:

$$\mathbf{F}_{ic_{21}} = -\mathbf{F}_{ic_{12}} = m_{3a} r\omega^2 \left(\cos\omega t\, \hat{\mathbf{i}} + \sin\omega t\, \hat{\mathbf{j}}\right) \tag{13.19b}$$

Figure 13-20b shows the free-body diagrams for the forces due only to the acceleration of the lumped mass of the crank at the crank pin, m_{2a}. These affect only the main pin at O_2. That component is:

$$\mathbf{F}_{ir} = -\mathbf{F}_{ir_{12}} = \mathbf{F}_{ir_{21}} = -m_{2a} \mathbf{a}_A$$

$$\mathbf{F}_{ir_{21}} = m_{2a} r\omega^2 \left(\cos\omega t\, \hat{\mathbf{i}} + \sin\omega t\, \hat{\mathbf{j}}\right) \tag{13.19c}$$

The gas force components were shown in Figure 13-7 and defined in equations 13.5.

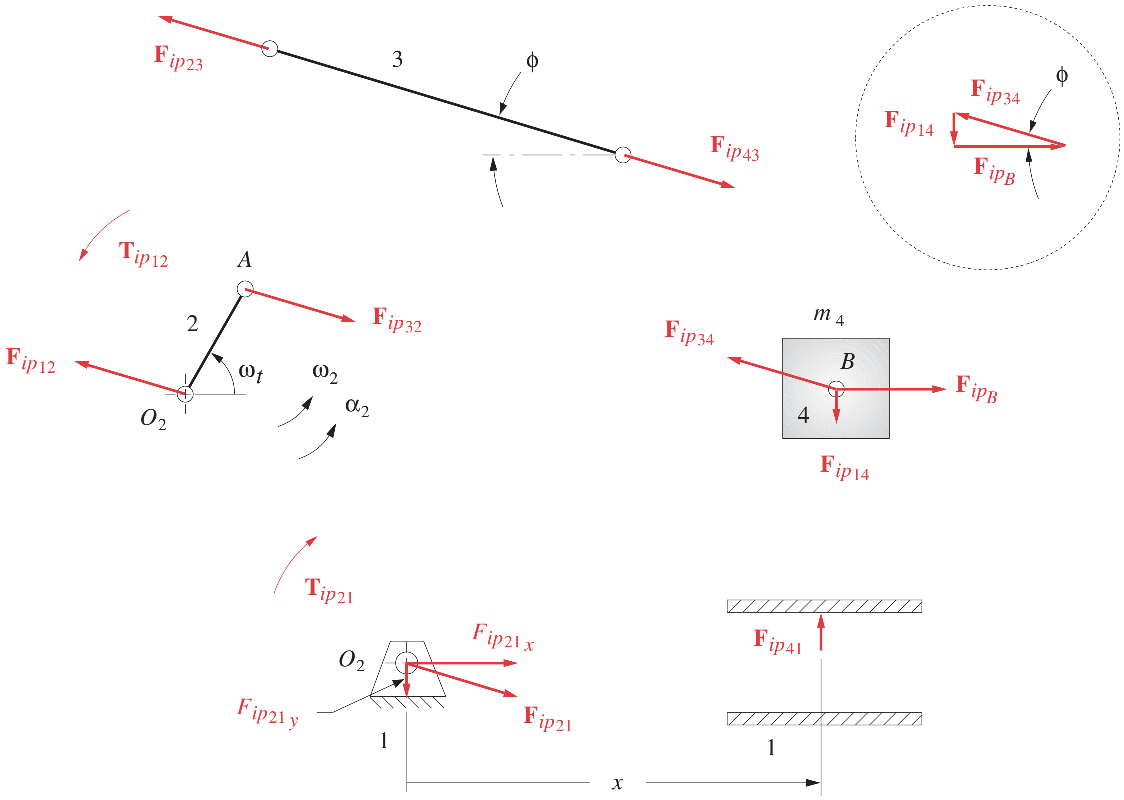


FIGURE 13-18

Free-body diagrams for forces due to the piston mass

We can now sum the components of the forces at each pin joint. For the sidewall force $\mathbf{F}_{41}$ of the piston against the cylinder wall:

$$\begin{aligned}\mathbf{F}_{41} &= \mathbf{F}_{g41} + \mathbf{F}_{ip41} + \mathbf{F}_{iw41} \\ &= -F_g \tan\phi\,\hat{\mathbf{j}} - m_4 a_B \tan\phi\,\hat{\mathbf{j}} - m_{3b} a_B \tan\phi\,\hat{\mathbf{j}} \\ &= -\left[\left(m_4 + m_{3b}\right)a_B + F_g\right]\tan\phi\,\hat{\mathbf{j}}\end{aligned} \tag{13.20}$$

The total force $\mathbf{F}_{34}$ on the wrist pin is:

$$\begin{aligned}\mathbf{F}_{34} &= \mathbf{F}_{g34} + \mathbf{F}_{ip34} + \mathbf{F}_{iw34} \\ &= \left(-F_g\,\hat{\mathbf{i}} - F_g \tan\phi\,\hat{\mathbf{j}}\right) + \left(m_4 a_B\,\hat{\mathbf{i}} - m_4 a_B \tan\phi\,\hat{\mathbf{j}}\right) + \left(-m_{3b} a_B \tan\phi\,\hat{\mathbf{j}}\right) \\ &= \left(-F_g + m_4 a_B\right)\hat{\mathbf{i}} - \left[F_g + \left(m_4 + m_{3b}\right)a_B\right]\tan\phi\,\hat{\mathbf{j}}\end{aligned} \tag{13.21}$$

The total force $\mathbf{F}_{32}$ on the crank pin is:

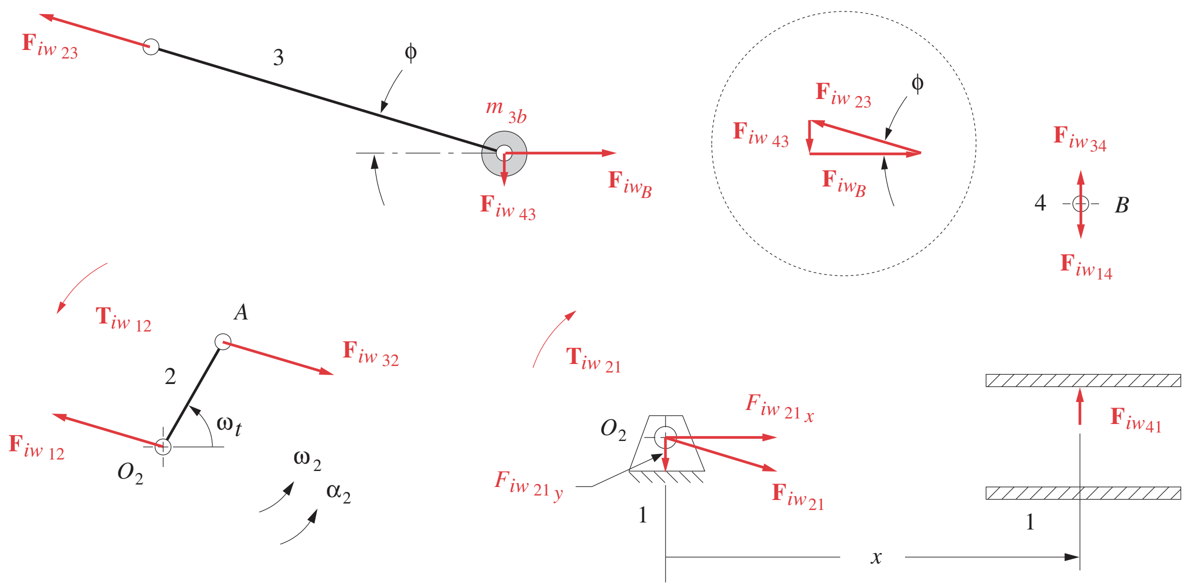
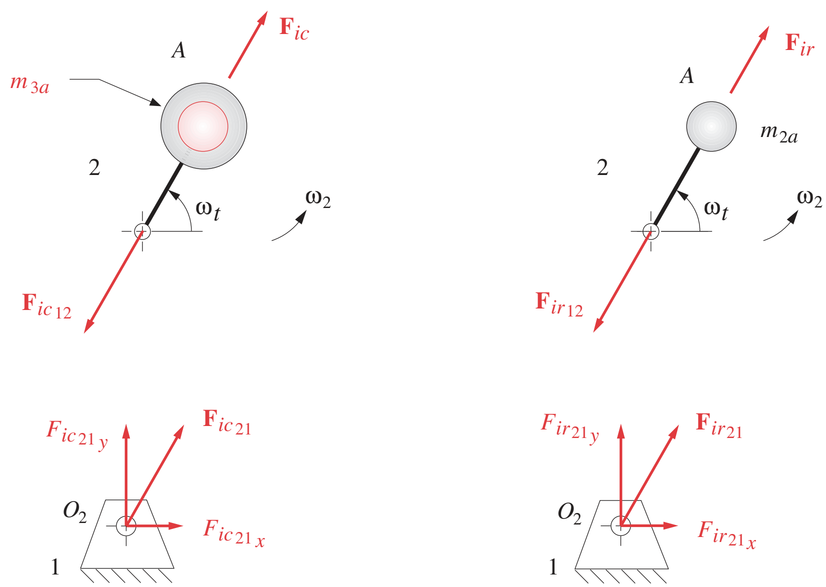


FIGURE 13-19

Free-body diagrams for forces due to the conrod mass concentrated at the wrist pin



(*a*) Conrod mass at crank pin (*b*) Crank mass at crank pin

FIGURE 13-20

Free-body diagrams for forces due to masses at the crank pin

$$\begin{aligned}\mathbf{F}_{32} &= \mathbf{F}_{g_{32}} + \mathbf{F}_{ip_{32}} + \mathbf{F}_{iw_{32}} + \mathbf{F}_{ic_{32}} \\ &= \left(F_g\,\hat{\mathbf{i}} + F_g \tan\phi\,\hat{\mathbf{j}}\right) + \left(-m_4 a_B\,\hat{\mathbf{i}} + m_4 a_B \tan\phi\,\hat{\mathbf{j}}\right) \\ &\qquad + \left(-m_{3b} a_B\,\hat{\mathbf{i}} + m_{3b} a_B \tan\phi\,\hat{\mathbf{j}}\right) + \left[m_{3a} r\omega^2\left(\cos\omega t\,\hat{\mathbf{i}} + \sin\omega t\,\hat{\mathbf{j}}\right)\right] \\ &= \left[m_{3a} r\omega^2 \cos\omega t - \left(m_{3b} + m_4\right) a_B + F_g\right]\hat{\mathbf{i}} \\ &\qquad + \left\{m_{3a} r\omega^2 \sin\omega t + \left[\left(m_{3b} + m_4\right) a_B + F_g\right]\tan\phi\right\}\hat{\mathbf{j}}\end{aligned} \tag{13.22}$$

The total force $\mathbf{F}_{21}$ on the main journal is:

$$\mathbf{F}_{21} = \mathbf{F}_{32} + \mathbf{F}_{ir_{21}} = \mathbf{F}_{32} + m_{2a} r\omega^2\left(\cos\omega t\,\hat{\mathbf{i}} + \sin\omega t\,\hat{\mathbf{j}}\right) \tag{13.23}$$

Note that, unlike the inertia force in equation 13.14, which was unaffected by the gas force, these pin forces **are** a function of the gas force as well as of the inertia forces. Engines with larger piston diameters will experience greater pin forces as a result of the explosion pressure acting on their larger piston area.

Program LINKAGES calculates the pin forces on all joints using equations 13.20 to 13.23. Figure 13-21 shows the wrist-pin force on the same unbalanced engine example as shown in previous figures, for three engine speeds. The “bow tie” loop is the inertia force and the “teardrop” loop is the gas force portion of the force curve. An interesting trade-off occurs between the gas force components and the inertia force components of the pin forces. At a low speed of 800 rpm (Figure 13-21a), the gas force dominates as the inertia forces are negligible at small ω. The peak wrist-pin force is then about 4200 lb. At high speed (6000 rpm), the inertia components dominate and the peak force is about 4500 lb (Figure 13-21c). But at a midrange speed (3400 rpm), the inertia force cancels some of the gas force and the peak force is only about 3200 lb (Figure 13-21b). These plots show that the pin forces can be quite large even in a moderately sized (0.4 liter/cylinder) engine. The pins, links, and bearings all have to be designed to withstand hundreds of millions of cycles of these reversing forces without failure.

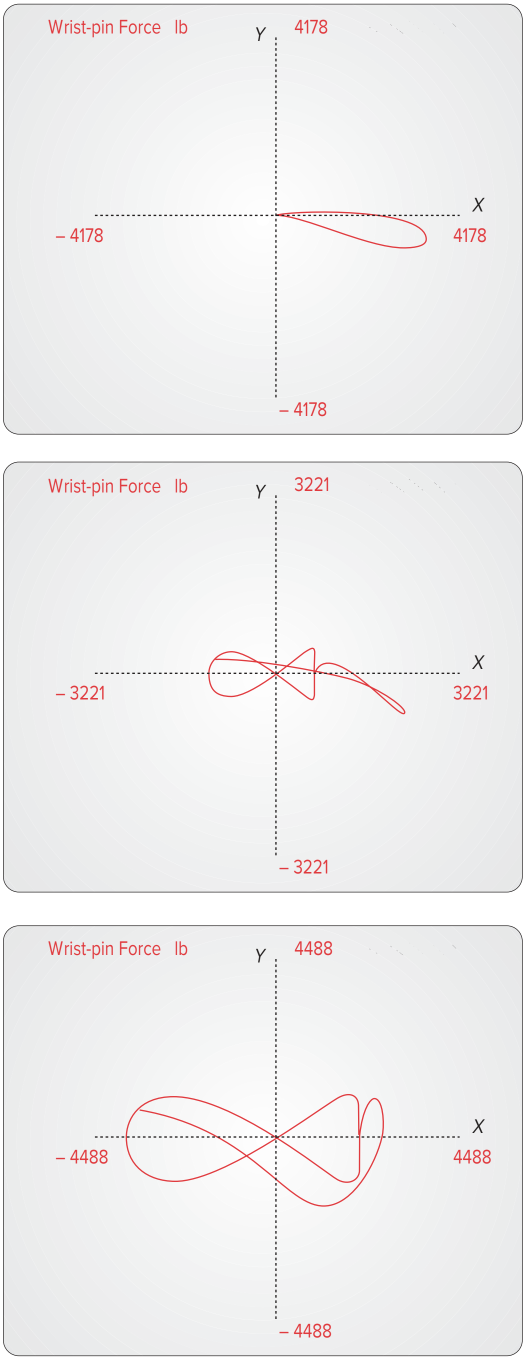
Figure 13-22 shows further evidence of the interaction of the gas forces and inertia forces on the crank pin and the wrist pin. Figures 13-22a and 13-22c show the variation in the magnitude of the inertia force component on the crank pin and wrist pin, respectively, over one full revolution of the crank as the engine speed is increased from idle to redline. Figures 13-22b and 13-22d show the variation in the total force on the same respective pins with both the inertia and gas force components included. These two plots show only the first 90° of crank revolution where the gas force in a four-stroke cylinder occurs. Note that the gas force and inertia force components counteract one another resulting in one particular speed where the total pin force is a minimum during the power stroke. This is the same phenomenon as seen in Figure 13-21.

Figure 13-23 shows the forces on the main pin and crank pin at three engine speeds for the same **unbalanced** single-cylinder engine example as in previous figures. These forces are plotted as hodographs in a local rotating coordinate system (LRCS) *x'y'* embedded in the crankshaft. Figure 13-23a shows that at 800 rpm (idle speed) the crank pin and main pin forces are essentially equal and opposite because the inertia force components are so small in comparison to the gas force components, which dominate at low speed. Only half the circumference of either pin sees any force. At 3400 rpm (Figure 13-23b),

(a) 800 rpm

(b) 3400 rpm

(c) 6000 rpm



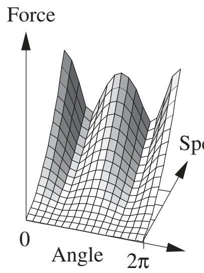
1 Cylinder
4 Stroke Cycle
Bore = 3.00 in
Stroke = 3.54
B/S = 0.85
L/R = 3.50
RPM = 3400
m_{2a} = 0.0150 bl
m_{3a} = 0.0134 bl
m_{3b} = 0.0066 bl
m_4 = 0.0050 bl
P_{max} = 600 psi

FIGURE 13-21

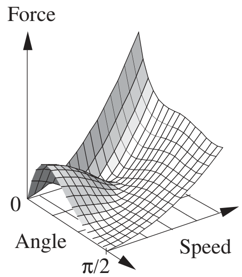
Forces on the wrist pin of the single-cylinder engine at various speeds the inertia force effects are evident and the angular portions of the main pin and crank pin that see any force are now only 39° and 72°, respectively. The effects of the gas force create asymmetry of the force hodographs about the *x'* axis. The differences between the main pin and crank pin forces are due to the different mass terms in their equations (e.g., compare equations 13.22 and 13.23).

In Figure 13-23c the engine is at redline (6000 rpm) and the inertia force components are now dominant, raising the peak force levels and making the hodographs nearly symmetrical about the *x*' axis. The angular portions of main pin and crank pin that see any force are now reduced to 30° and 54°, respectively. This force distribution causes crank pins to wear only on a portion of their circumference. As shown in the next section, crank balancing affects the force distribution on the main pins.

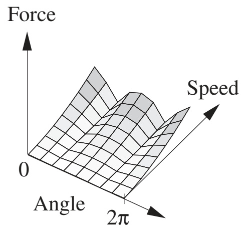
Note that the numerical values of force and torque in the figures of this chapter are unique to the arbitrary choice of engine parameters used for their example engine and should not be extrapolated to any other engine design. Also, the gas force function used in program LINKAGES to generate the figures is both approximate and invariant with engine speed, unlike in a real engine. Use the equations of this chapter to calculate forces and torques using mass, geometry, and gas force data appropriate to your particular engine design.



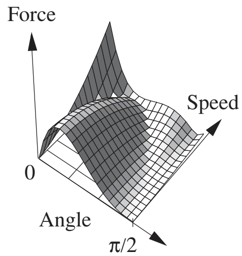
(*a*) Crank-pin inertia force



(*b*) Crank-pin total force



(*c*) Wrist-pin inertia force



(*d*) Wrist-pin total force

FIGURE 13-22
Pin force variation

13.10 BALANCING THE SINGLE-CYLINDER ENGINE *Watch a Short Video on Engine Balancing (21:46)**

The derivations and figures in the preceding sections have shown that significant forces are developed both on the pivot pins and on the ground plane due to the gas forces and the inertia and shaking forces. Balancing will not have any effect on the gas forces, which are internal, but it can have a dramatic effect on the inertia and shaking forces. The main pin force can be reduced, but the crank-pin and wrist-pin forces will be unaffected by any crankshaft balancing done. Figure 13-13 shows the unbalanced shaking force as felt on the ground plane of our 0.4-liter single-cylinder example engine from program LINKAGES. It is about 9700 lb even at the moderate speed of 3400 rpm. At 6000 rpm it increases to over 30 000 lb. The methods of Chapter 12 can be applied to this mechanism to balance the members in pure rotation and reduce these large shaking forces.

Figure 13-24a shows the dynamic model of our slider-crank with the conrod mass lumped at both crank pin *A* and wrist pin *B*. We can consider this single-cylinder engine to be a single-plane device, thus suitable for static balancing (see Section 13.1). It is straightforward to statically balance the crank. We need a balance mass at some radius, 180° from the lumped mass at point *A* whose *mr* product is equal to the product of the mass at *A* and its radius *r*. Applying equation 12.2 to this simple problem, we get:

$$m_{bal}\mathbf{R}_{bal} = -m_A\mathbf{R}_A \tag{13.24}$$

Any combination of mass and radius that gives this product, placed at 180° from point *A,* will balance the crank. For simplicity of example, we will use a balance radius equal to *r*. Then a mass equal to m_A placed at *A'* will exactly balance the rotating masses. The *CG* of the crank will then be at the fixed pivot O_2 as shown in Figure 13-24a. In a real crankshaft, actually placing the counterweight *CG* at this large a radius will not work.

* http://www.designofmachinery.com/DOM/Balancing_One_Cylinder.mp4

13

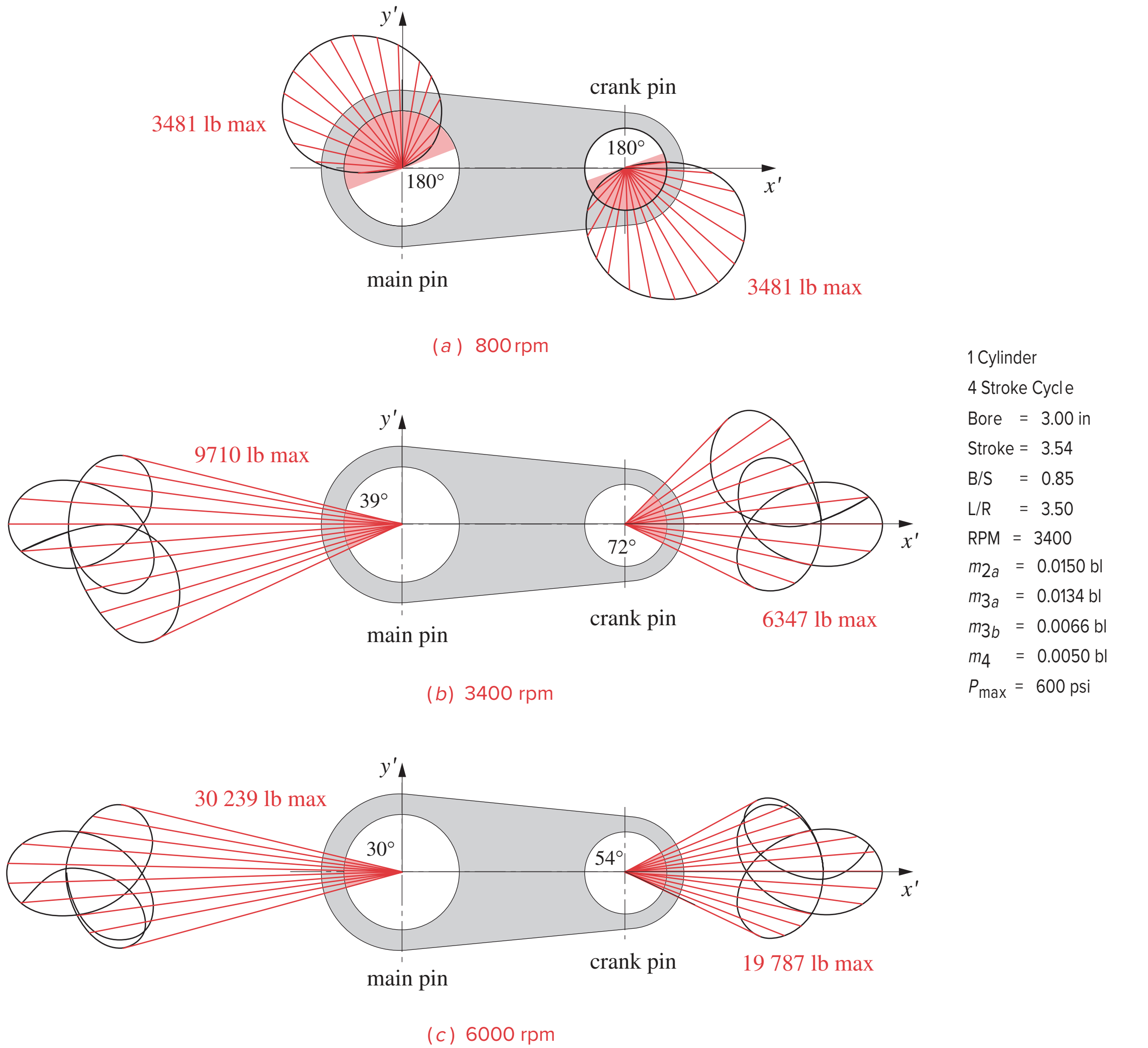
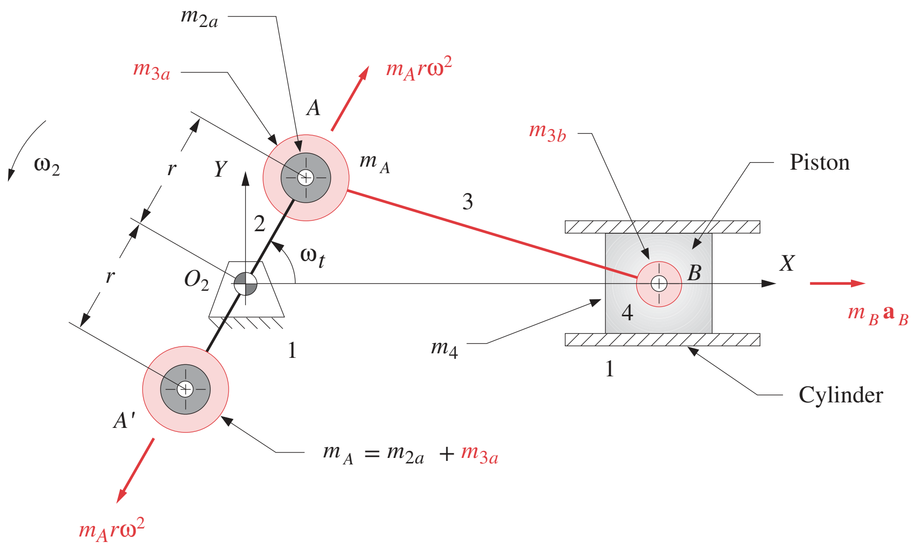


FIGURE 13-23

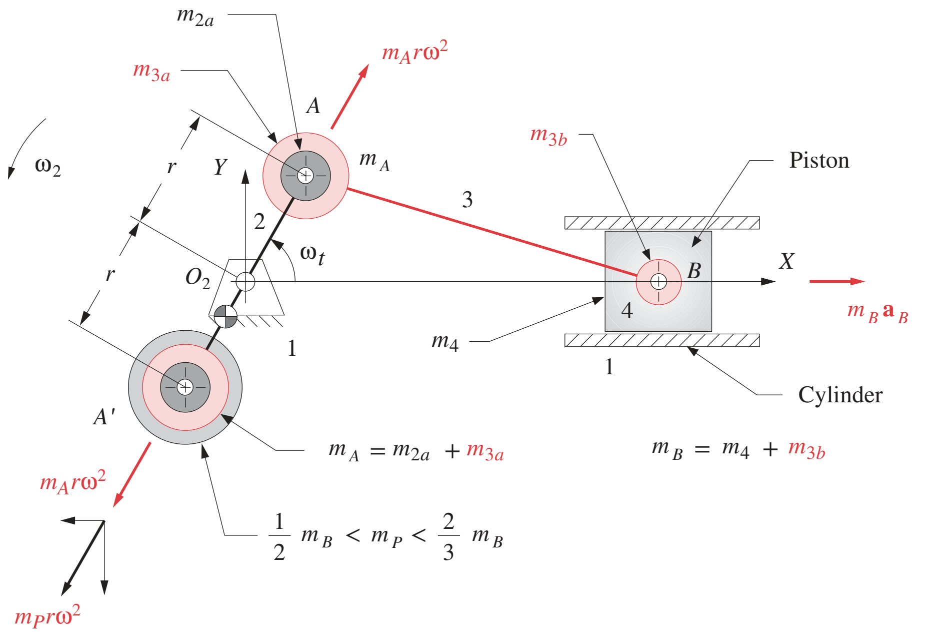
Hodographs of dynamic forces on main pin and crank pin of an unbalanced, one-cylinder, four-stroke engine running at various speeds

The balance mass has to be kept close to the centerline to clear the piston at BDC. Figure 13-2c shows the shape of typical crankshaft counterweights.

Figure 13-25a shows the shaking force from the same engine as in Figure 13-13 after the crank has been exactly balanced in this manner. The Y component of the shaking force has been reduced to zero and the X component to 3343 lb at 3400 rpm. This is a factor of three reduction over the unbalanced engine. Note that the only source of Y-directed inertia force is the rotating mass at point A of Figure 13-24 (see equations 13.14). What remains after balancing the rotating mass is the force due to the acceleration of the piston



(a) Crank exactly balanced



(b) Crank overbalanced

FIGURE 13-24

Balancing and overbalancing the single-cylinder engine

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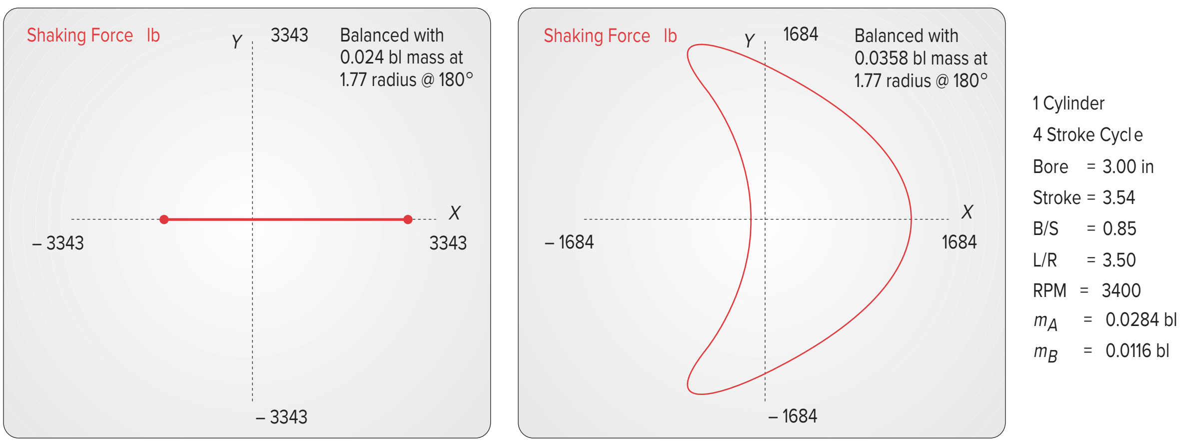
and conrod masses at point B of Figure 13-24 which are in linear translation along the X axis, as shown by the inertia force $-m_B\mathbf{a}_B$ at point B in that figure.

To completely eliminate this reciprocating unbalanced shaking force would require the introduction of another reciprocating mass, which oscillates 180° out of phase with the piston. Adding a second piston and cylinder, properly arranged, can accomplish this. One of the principal advantages of multicylinder engines is their ability to reduce or eliminate the shaking forces. We will investigate this in the next chapter.

In the single-cylinder engine, there is no way to completely eliminate the reciprocating unbalance with a single, rotating counterweight, but we can reduce the shaking force still further. Figure 13-24b shows an additional amount of mass m_P added to the counterweight at point *A*'. (Note that the crank's *CG* has now moved away from the fixed pivot.) This extra balance mass creates an additional inertia force $(-m_P r\omega^2)$ as is shown, broken into *X* and *Y* components, in the figure. The *Y* component is not opposed by any other inertia forces present, but the *X* component will always be opposite to the reciprocating inertia force at point *B*. Thus this extra mass, m_P, which *overbalances the crank*, will reduce the *X*-directed shaking force at the expense of adding back some *Y*-directed shaking force. This is a useful trade-off as the direction of the shaking force is usually of less concern than is its magnitude. Shaking forces create vibrations in the supporting structure which are transmitted through it and modified by it. As an example, it is unlikely that you could define the direction of a motorcycle engine's shaking forces by feeling their resultant vibrations in the handlebars. But you **will** detect an increase in the magnitude of the shaking forces from the larger amplitude of vibration they cause in the cycle frame.

The correct amount of additional "overbalance" mass needed to minimize the peak shaking force, regardless of its direction, will vary with the particular engine design. It will usually be between one-half and two-thirds of the reciprocating mass at point *B* (piston plus conrod at wrist pin), if placed at the crank radius *r*. Of course, once this mass-radius product is determined, it can be achieved with any combination of mass and radius. Figure 13-25b shows the minimum shaking force achieved for this engine with the addition of 65.5% of the mass at *B* acting at radius *r*. The shaking force has now been



(*a*) Crank exactly balanced (*b*) Crank overbalanced

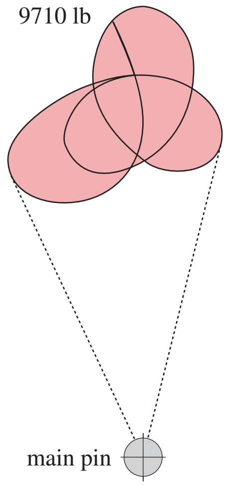
FIGURE 13-25

Effects of balancing and overbalancing on the shaking force in the slider-crank linkage reduced to 1684 lb at 3400 rpm, which is **17% of its original unbalanced value** of 9710 lb. The benefits of balancing, and of overbalancing in the case of the single-cylinder engine, should now be obvious.

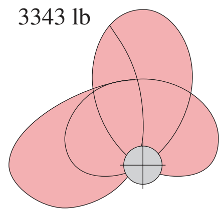
Effect of Crankshaft Balancing on Pin Forces

Of the pin forces, only the main pin force is affected by the addition of balance mass to the crankshaft. This is because its equation (13.23) is the only one of the pin-force equations (13.20 to 13.23) that contains the mass of the crank. Table 13-3 shows the magnitudes of the shaking forces (in GCS) and main pin forces (in LCRS) for the single-cylinder example engine of Figure 13-23 at three engine speeds and under three conditions of balance: unbalanced, exactly balanced with a counterweight mass equal to the total mass m_A at the crank pin (Figure 13-25a), and overbalanced with the mass needed to minimize the single-cylinder shaking force (Figure 13-25b). Note that both balancing and overbalancing reduce the main pin force, though to a lesser degree than they reduce the shaking force in some cases. At idle speed the gas force far exceeds the inertia force and, since balancing can only affect the latter, the reduction in main pin force is less at idle speed than at higher engine speeds. The main pin forces in the overbalanced case most closely track the shaking forces at the redline speed where inertia force dominates gas force. Note that overbalancing the crank reduces the main pin force below that of the exact balancing case at all speeds.*

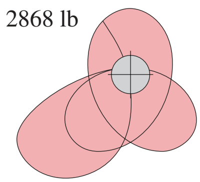
Figure 13-26 shows the effect of balancing and overbalancing on the main pin force magnitude and its distribution. Not only is the peak unbalanced main pin force (Figure 13-26a) nearly 3 times the magnitude of the exactly balanced case (Figure 13-26b) but the forces in the unbalanced case are concentrated over a small portion of the pin circumference. (See also Figure 13-23.) The exactly balanced crankshaft has its main pin force distributed over more than half its circumference and the overbalanced crankshaft puts the force completely around the pin circumference as shown in Figure 13-26c.



(*a*) Unbalanced



(*b*) Exact balance



(*c*) Overbalanced

FIGURE 13-26

Force on main pin at 3400 rpm with different crank balance states, shown at same scale in local rotating coordinates (LRCS)

13.11 DESIGN TRADE-OFFS AND RATIOS

In the design of any system or device, no matter how simple, there will always be conflicting demands, requirements, or desires which must be traded off to achieve the best design compromise. This single-cylinder engine is no exception. There are two dimensionless design ratios which can be used to characterize an engine's dynamic behavior in a general way. The first is the crank/conrod ratio r/l, introduced in Section 13.2, or its inverse, the **conrod/crank ratio** l/r. The second is the **bore/stroke ratio** B/S.

TABLE 13-3 Effect of Crank Balance Mass on Shaking Force and Mainpin Force

Balance Mode	Peak Shaking Force Mag. (lb)			Peak Mainpin Force Mag. (lb)		
	Idle	Midrange	Redline	Idle	Midrange	Redline
Unbalanced	538	9710	30 239	3481	9710	30 239
Exact balance	185	3343	10 412	4095	3343	10 412
Overbalanced	33	1684	5246	3675	2868	5886

Conrod/Crank Ratio

The crank/conrod ratio r/l appears in all the equations for acceleration, forces, and torques. In general, the smaller the r/l ratio, the smoother will be the acceleration function and thus all other factors which it influences. Program LINKAGES uses the inverse of this ratio as an input parameter. The **conrod/crank ratio** l/r must be greater than about two to obtain acceptable transmission angles in the slider-crank linkage. The ideal value for l/r from a kinematic standpoint would be infinity as that would result in the piston acceleration function being a pure harmonic. The second and all subsequent harmonic terms in equations 13.3 would be zero in this event, and the peak value of acceleration would be a minimum. However, an engine this tall would not package very well, and package considerations often dictate the maximum value of the l/r ratio. Most engines will have an l/r ratio between three and five which values give acceptable smoothness in a reasonably short engine.

Bore/Stroke Ratio

The bore B of the cylinder is essentially equal to the diameter of the piston. (There is a small clearance.) The stroke S is defined as the distance travelled by the piston from TDC to BDC and is twice the crank radius, $S = 2r$. The bore appears in the equation for gas force (equation 13.4) and thus also affects gas torque. The crank radius appears in every equation. An engine with a B/S ratio of 1 is referred to as a “square” engine. If B/S is larger than 1, it is “oversquare”; if less, “undersquare.” The designer’s choice of this ratio can have a significant effect on the dynamic behavior of the engine. Assuming that the displacement, or stroke volume V, of the engine has been chosen and is to remain constant, this displacement can be achieved with an infinity of combinations of bore and stroke ranging from a “pancake” piston with tiny stroke to a “pencil” piston with very long stroke.

$$V = \frac{\pi B^2}{4} S \tag{13.25}$$

There is a classic design trade-off here between B and S for a constant stroke volume V. A large bore and small stroke will result in high gas forces which will affect pin forces adversely. A large stroke and small bore will result in high inertia forces which will affect pin forces (as well as other forces and torques) adversely. So there should be an optimum value for the B/S ratio in each case, which will minimize these adverse effects. Most production engines have B/S ratios in the range of about 0.75 to 1.5.

It is left as an exercise for the reader to investigate the effects of variation in the B/S and l/r ratios on forces and torques in the system. Program LINKAGES will demonstrate the effects of changes made independently to each of these ratios, while all other design parameters are held constant. The reader is encouraged to experiment with the program to gain insight into the role of these ratios in the dynamic performance of the engine.

Materials

There will always be a strength/weight trade-off. The forces in this device can be quite high, due both to the explosion and to the inertia of the moving elements. We would like to keep the masses of the parts as low as possible as the accelerations are typically very high, as seen in Figure 13-8c. But the parts must be strong enough to withstand the forces,

* (From previous page) Overbalancing a 4-cylinder inline engine that uses eight balance masses (two per cylinder split on either side of each crank throw) with 100% of $m_A R_A$ plus 50% of $m_B R_A$ per cylinder will minimize its main bearing forces. If four crankshaft counterbalance weights are used (one per cylinder on one side of each crank throw in a particular arrangement), then the optimum balance condition to minimize main bearing forces is 67% of $m_A R_A$ plus 33% of $m_B R_A$ per cylinder. (*Source: Chrysler Corp.*).

so materials with good strength-to-weight ratios are needed. Pistons are usually made of an aluminum alloy, either cast or forged. Conrods are most often cast ductile iron or forged steel, except in very small engines (lawn mower, chain saw, motorcycle) where they may be aluminum alloy. Some high-performance engines (e.g. Acura NSX) have titanium connecting rods. Crankshafts are usually forged steel or cast ductile iron, and wrist pins are of hardened steel tubing or rod. Plain bearings of a special soft, nonferrous metal alloy called babbitt are usually used. In the four-stroke engine these are pressure lubricated with oil pumped through drilled passageways in the block, crankshaft, and connecting rods. In the two-stroke engine, the fuel carries the lubricant to these parts. Engine blocks are cast iron or cast aluminum alloy. The chrome-plated steel piston rings seal and wear well against gray cast iron cylinders. Most aluminum blocks are fitted with cast iron liners around the cylinder bores. Some are unlined and made of a high-silicon aluminum alloy which is specially cooled after casting to precipitate the hard silicon in the cylinder walls for wear resistance.

13.12 BIBLIOGRAPHY

1 **Heywood, J. B.** (1988). *Internal Combustion Engine Fundamentals.* McGraw-Hill: New York.

2 **Taylor, C. F.** (1966). *The Internal Combustion Engine in Theory and Practice.* MIT Press: Cambridge, MA.

3 **Heisler, H.** (1999). *Vehicle and Engine Technology, 2ed.* SAE: Warrendale, PA.

13.13 PROBLEMS‡

*†13-1 A slider-crank linkage has $r = 3$ and $l = 12$, $\omega = 200$ rad/sec at time $t = 0$. Its initial crank angle is zero. Calculate the piston acceleration at $t = 1$ sec. Use two methods, the exact solution and the approximate Fourier series solution, and compare the results.

†13-2 Repeat Problem 13-1 for $r = 4$ and $l = 15$ and $t = 0.9$ sec.

*†13-3 A slider-crank linkage has $r = 3$ and $l = 12$, and a piston bore $B = 2$. The peak gas pressure in the cylinder occurs at a crank angle of 10° and is 1000 pressure units. Calculate the gas force and gas torque at this position.

†13-4 A slider-crank linkage has $r = 4$ and $l = 15$, and a piston bore $B = 3$. The peak gas pressure in the cylinder occurs at a crank angle of 5° and is 600 pressure units. Calculate the gas force and gas torque at this position.

*†13-5 Repeat Problem 13-3 using the exact method of calculation of gas torque and compare its result to that obtained by the approximate equation 13.8b. What is the % error?

†13-6 Repeat Problem 13-4 using the exact method of calculation of gas torque and compare its result to that obtained by the approximate equation 13.8b. What is the % error?

*†13-7 A connecting rod of length $l = 12$ has a mass $m_3 = 0.020$.§ Its mass moment of inertia is 0.620. Its *CG* is located at $0.4l$ from the crank pin, point *A*.

a. Calculate an exact dynamic model using two lumped masses, one at the wrist pin, point *B*, and one at whatever other point is required. Define the lumped masses and their locations.

b. Calculate an approximate dynamic model using two lumped masses, one at the wrist

§ Note that, unless otherwise stated, all masses are expressed in lb-sec^2 inertia in bl-in^2.

TABLE P13-0
Topic/Problem Matrix

13.2 Slider-Crank Kinematics
13-1, 13-2, 13-34, 13-35, 13-36, 13-37, 13-59

13.3 Gas Force and Gas Torque
13-3, 13-4, 13-5, 13-6, 13-38, 13-39, 13-40, 13-41, 13-42, 13-60

13.4 Equivalent Masses
13-7, 13-8, 13-9, 13-10, 13-43, 13-44, 13-45, 13-46, 13-61

13.6 Inertia and Shaking Torques
13-11, 13-12, 13-13, 13-14, 13-47, 13-48, 13-49, 13-50, 13-62

13.9 Pin Forces
13-15, 13-16, 13-17, 13-18, 13-23, 13-24, 13-25, 13-26, 13-27, 13-28, 13-33, 13-51, 13-52, 13-53, 13-54, 13-63

13.10 Balancing the Single-Cylinder Engine
13-19, 13-20, 13-21, 13-22, 13-29, 13-30, 13-31, 13-32, 13-55, 13-56, 13-57, 13-58, 13-64

‡ All problem figures are provided as PDF files, and some are also provided as animated Working Model files; all are downloadable. PDF filenames are the same as the figure number.

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

pin, point B, and one at the crank pin, point A. Define the lumped masses and their locations.

c. Calculate the error in the mass moment of inertia of the approximate model as a percentage of the original mass moment of inertia.

†13-8 Repeat Problem 13-7 for these data: $l = 15$, mass $m_3 = 0.025$,§ mass moment of inertia is 1.020. Its CG is located at $0.25l$ from the crank pin, point A.

*†13-9 A crank of length $r = 3.5$ has a mass $m_2 = 0.060$.§ Its mass moment of inertia about its pivot is 0.300. Its CG is at $0.30r$ from the main pin, O_2. Calculate a statically equivalent two-lumped mass dynamic model with the lumps placed at the main pin and crank pin. What is the percent error in the model's moment of inertia about the crank pivot?

†13-10 Repeat Problem 13-9 for a crank length $r = 4$, a mass $m_2 = 0.050$,§ a mass moment of inertia about its pivot of 0.400. Its CG is located at $0.40r$ from the main pin, point O_2.

*†13-11 Combine the data from Problems 13-7 and 13-9. Run the linkage at a constant 2000 rpm. Calculate the inertia force and inertia torque at $\omega t = 45°$. Piston mass = 0.012.§

†13-12 Combine the data from Problems 13-7 and 13-10. Run the linkage at a constant 3000 rpm. Calculate the inertia force and inertia torque at $\omega t = 30°$. Piston mass = 0.019.§

†13-13 Combine the data from Problems 13-8 and 13-9. Run the linkage at a constant 2500 rpm. Calculate the inertia force and inertia torque at $\omega t = 24°$. Piston mass = 0.023.§

*†13-14 Combine the data from Problems 13-8 and 13-10. Run the linkage at a constant 4000 rpm. Calculate the inertia force and inertia torque at $\omega t = 18°$. Piston mass = 0.015.§

†13-15 Combine the data from Problems 13-7 and 13-9. Run the linkage at a constant 2000 rpm. Calculate the pin forces at $\omega t = 45°$. Piston mass = 0.022.§ $F_g = 300$.

†13-16 Combine the data from Problems 13-7 and 13-10. Run the linkage at a constant 3000 rpm. Calculate the pin forces at $\omega t = 30°$. Piston mass = 0.019.§ $F_g = 600$.

†13-17 Combine the data from Problems 13-8 and 13-9. Run the linkage at a constant 2500 rpm. Calculate the pin forces at $\omega t = 24°$. Piston mass = 0.032.§ $F_g = 900$.

†13-18 Combine the data from Problems 13-8 and 13-10. Run the linkage at a constant 4000 rpm. Calculate the pin forces at $\omega t = 18°$. Piston mass = 0.014.§ $F_g = 1200$.

*†‡13-19 Using the data from Problem 13-11:

a. Exactly balance the crank and recalculate the inertia force.
b. Overbalance the crank with approximately two-thirds of the mass at the wrist pin placed at radius $-r$ on the crank and recalculate the inertia force.
c. Compare these results to those for the unbalanced crank.

†‡13-20 Repeat Problem 13-19 using the data from Problem 13-12.

†‡13-21 Repeat Problem 13-19 using the data from Problem 13-13.

†‡13-22 Repeat Problem 13-19 using the data from Problem 13-14.

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program LINKAGES.

§ Note that, unless otherwise stated, all masses are expressed in lb-sec²/in or blobs (bl) and mass moments of inertia in bl-in².

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program LINKAGES.

†13-23 Combine the necessary equations to develop expressions that show how each of these dynamic parameters varies as a function of the crank/conrod ratio alone:

a. Piston acceleration b. Inertia force
c. Inertia torque d. Pin forces

Plot the functions. Check your conclusions with program LINKAGES.

Hint: *Consider all other parameters to be temporarily constant. Set the crank angle to some value such that the gas force is nonzero.*

†13-24 Combine the necessary equations to develop expressions that show how each of these dynamic parameters varies as a function of the bore/stroke ratio alone:

a. Gas force b. Gas torque c. Inertia force
d. Inertia torque e. Pin forces

Plot the functions. Check your conclusions with program LINKAGES.

Hint: *Consider all other parameters to be temporarily constant. Set the crank angle to some value such that the gas force is nonzero.*

†13-25 Develop an expression to determine the optimum bore/stroke ratio to minimize the wrist-pin force. Plot the function.

†‡13-26 Use program LINKAGES, your own computer program, or an equation solver to calculate the maximum value and the polar-plot shape of the force on the main pin of a 1-in^3 displacement, single-cylinder engine with bore = 1.12838 in for the following situations:

a. Piston, conrod, and crank masses = 0
b. Piston mass = 1 blob, conrod and crank masses = 0
c. Conrod mass = 1 blob, piston and crank masses = 0
d. Crank mass = 1 blob, conrod and piston masses = 0

Place the *CG* of the crank at 0.5r and the conrod at 0.33l. Compare and explain the differences in the main pin force under these different conditions with reference to the governing equations.

†13-27 Repeat Problem 13-26 for the crank pin.

†13-28 Repeat Problem 13-26 for the wrist pin.

†‡13-29 Use program LINKAGES, your own computer program, or an equation solver to calculate the maximum value and the polar-plot shape of the force on the main pin of a 1-in^3 single-cylinder engine with bore = 1.12838 in for the following situations:

a. Engine unbalanced.
b. Crank exactly balanced against mass at crank pin.
c. Crank optimally overbalanced against masses at crank pin and wrist pin.

Piston, conrod, and crank masses = 1. Place the *CG* of the crank at 0.5r and the conrod at 0.33l. Compare and explain the differences in the main pin force under these conditions with reference to the governing equations.

†‡13-30 Repeat Problem 13-29 for the crankpin force.

†‡13-31 Repeat Problem 13-29 for the wrist pin force.

†‡13-32 Repeat Problem 13-29 for the shaking force.

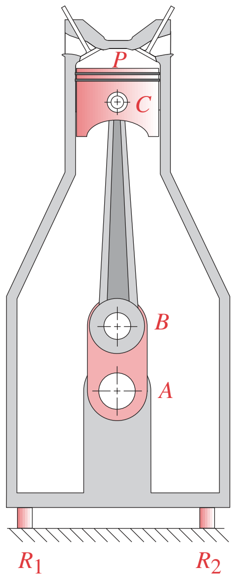


FIGURE P13-1
Problem 13-33

†13-33 Figure P13-1 shows a single-cylinder air compressor stopped at top dead center (TDC). There is a static pressure $P = 100$ psi trapped in the 3-in-bore cylinder. The entire assembly weighs 30 lb. Draw the necessary free-body diagrams to determine the forces at points A, B, C, and the supports R_1 and R_2, which are symmetrically located about the piston centerline. Assume that the piston remains stationary.

†13-34 Calculate and plot the position, velocity, and acceleration of a slider-crank linkage with $r = 3$, $l = 12$, and $\omega = 200$ rad/sec over one cycle using the exact solution and the approximate Fourier series solution. Also, calculate and plot the percent difference between the exact and approximate solutions for acceleration.

†13-35 Repeat Problem 13-34 for $r = 3$, $l = 15$, and $\omega = 100$ rad/sec.

†13-36 A slider-crank linkage has $r = 3$, $l = 9$. It has an angular velocity of 100 rad/sec at time $t = 0$. Its initial crank angle is zero. Calculate the piston acceleration at $t = 0.01$ sec. Compare the exact solution to the approximate Fourier series solution.

†13-37 Repeat Problem 13-36 with $r = 3$, $l = 15$, and $t = 0.02$.

†13-38 The following equation is an approximation of the gas force over 180° of crank angle.

$$F_g = \begin{cases} F_{g_{max}} \sin\left[(\omega t/\beta)(\pi/2)\right], & 0 \le \omega t \le \beta \\ F_{g_{max}} \left\{1 + \cos\left[\pi(\omega t - \beta)/(\pi - \beta)\right]\right\}/2, & \beta < \omega t \le \pi \end{cases}$$

Using this equation with $\beta = 15°$ and $F_{g_{max}} = 1200$ lb, calculate and plot the approximate gas torque for $r = 4$ in and $l = 12$ in. What is the total energy delivered by the gas force over the 180° of motion? What is the average power delivered if the crank rotates at a constant speed of 1500 rpm?

†13-39 A slider-crank linkage has $r = 3.75$, $l = 11$ and a piston bore of $B = 2.5$. The peak gas pressure in the cylinder occurs at a crank angle of 12° and is 1150 pressure units. Calculate the gas force and gas torque at this position.

†13-40 Repeat Problem 13-39 using the exact method of calculation of the gas torque and compare the result to that obtained by the approximate expression in equation 13.8b. What is the percent error?

†13-41 A slider-crank linkage has $r = 4.12$, $l = 14.5$ and a piston bore of $B = 2.25$. The peak gas pressure in the cylinder occurs at a crank angle of 9° and is 1325 pressure units. Calculate the gas force and gas torque at this position.

†13-42 Repeat Problem 13-41 using the exact method of calculation of the gas torque and compare the result to that obtained by the approximate expression in equation 13.8b. What is the percent error?

†13-43 A slider-crank linkage has crank $m_2 = 0.045$,§ crank $r_{G_2} = 0.4r$, conrod $m_3 = 0.12$,§ and piston $m_4 = 0.15$. Determine the approximately dynamically equivalent two-mass lumped parameter model for this linkage with the masses placed at the crank and wrist pins.

†13-44 If the conrod in Problem 13-43 has $l = 12.5$, $I_{G_3} = 0.15$§ and its CG is located 4.5 units from point A; calculate the sizes of two dynamically equivalent masses and the location of one if the other is placed at point B (see Figure 13-10).

†13-45 A slider-crank linkage has crank $m_2 = 0.060$,§ crank $r_{G_2} = 0.38r$, conrod $m_3 = 0.18$, and piston $m_4 = 0.16$. Determine the approximately dynamically equivalent two-mass lumped parameter model for this linkage with the masses placed at the crank and wrist pins.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

§ Note that, unless otherwise stated, all masses are expressed in lb-sec^2/in or blobs (bl) and mass moments of inertia in bl-in^2.

†13-46 If the conrod in Problem 13-45 has $l = 10.4$, $I_{G_3} = 0.12$§ and its *CG* is located 4.16 units from point *A*; calculate the sizes of two dynamically equivalent masses and the location of one if the other is placed at point *B* (see Figure 13-10).

†13-47 A slider-crank linkage has $r = 3.13$, $l = 12.5$, crank $m_2 = 0.045$,§ crank $r_{G_2} = 0.4r$, conrod $m_3 = 0.12$, conrod $r_{G_3} = 0.36l$, and piston $m_4 = 0.15$. Crank $\omega = 1800$ rpm. Calculate the inertia force and inertia torque for a crank position of $\omega t = 30°$.

†13-48 A slider-crank linkage has $r = 2.6$, $l = 10.4$, crank $m_2 = 0.060$,§ crank $r_{G_2} = 0.38r$, conrod $m_3 = 0.18$, conrod $r_{G_3} = 0.4l$, and piston $m_4 = 0.16$. Crank $\omega = 1850$ rpm. Calculate the inertia force and inertia torque for a crank position of $\omega t = 20°$.

†13-49 A slider-crank linkage has $r = 2.6$, $l = 10.4$, crank $m_2 = 0.045$,§ crank $r_{G_2} = 0.4r$, conrod $m_3 = 0.12$, conrod $r_{G_3} = 0.36l$, and piston $m_4 = 0.15$. Crank $\omega = 2000$ rpm. Calculate the inertia force and inertia torque for a crank position of $\omega t = 25°$.

†13-50 A slider-crank linkage has $r = 3.13$, $l = 12.5$, crank $m_2 = 0.060$,§ crank $r_{G_2} = 0.38r$, conrod $m_3 = 0.18$, conrod $r_{G_3} = 0.4l$, and piston $m_4 = 0.15$. Crank $\omega = 1500$ rpm. Calculate the inertia force and inertia torque for a crank position of $\omega t = 22°$.

†13-51 A slider-crank linkage has $r = 3.13$, $l = 12.5$, crank $m_2 = 0.045$,§ crank $r_{G_2} = 0.4r$, conrod $m_3 = 0.12$, conrod $r_{G_3} = 0.36l$, and piston $m_4 = 0.15$. Crank $\omega = 1800$ rpm. Calculate the pin forces for a crank position of $\omega t = 30°$ and a gas force of $F_g = 450$.

†13-52 A slider-crank linkage has $r = 2.6$, $l = 10.4$, crank $m_2 = 0.060$,§ crank $r_{G_2} = 0.38r$, conrod $m_3 = 0.18$, conrod $r_{G_3} = 0.4l$, and piston $m_4 = 0.16$. Crank $\omega = 1850$ rpm. Calculate the pin forces for a crank position of $\omega t = 20°$ and a gas force of $F_g = 600$.

†13-53 A slider-crank linkage has $r = 2.6$, $l = 10.4$, crank $m_2 = 0.045$,§ crank $r_{G_2} = 0.4r$, conrod $m_3 = 0.12$, conrod $r_{G_3} = 0.36l$, and piston $m_4 = 0.15$. Crank $\omega = 2000$ rpm. Calculate the pin forces for a crank position of $\omega t = 25°$ and a gas force of $F_g = 350$.

†13-54 A slider-crank linkage has $r = 3.13$, $l = 12.5$, crank $m_2 = 0.060$,§ crank $r_{G_2} = 0.38r$, conrod $m_3 = 0.18$, conrod $r_{G_3} = 0.4l$, and piston $m_4 = 0.15$. Crank $\omega = 1500$ rpm. Calculate the pin forces for a crank position of $\omega t = 22°$ and a gas force of $F_g = 550$.

†‡13-55 Using the data from Problem 13-47:

a. Exactly balance the crank and recalculate the inertia force.
b. Overbalance the crank with approximately two-thirds of the mass at the wrist pin placed at radius $-r$ on the crank and recalculate the inertia force.
c. Compare these results to those for the unbalanced crank.

†‡13-56 Using the data from Problem 13-48:

a. Exactly balance the crank and recalculate the inertia force.
b. Overbalance the crank with approximately two-thirds of the mass at the wrist pin placed at radius $-r$ on the crank and recalculate the inertia force.
c. Compare these results to those for the unbalanced crank.

†‡13-57 Using the data from Problem 13-49:

a. Exactly balance the crank and recalculate the inertia force.
b. Overbalance the crank with approximately two-thirds of the mass at the wrist pin placed at radius $-r$ on the crank and recalculate the inertia force.
c. Compare these results to those for the unbalanced crank.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program LINKAGES.

§ Note that, unless otherwise stated, all masses are expressed in lb-sec^2/in or blobs (bl) and mass moments of inertia in bl-in^2.

† These problems are suited to solution using *Mathcad, Matlab,* or *TKSolver* equation solver programs.

‡ These problems and projects are suited to solution using program LINKAGES.

†‡13-58 Using the data from Problem 13-50:

a. Exactly balance the crank and recalculate the inertia force.
b. Overbalance the crank with approximately two-thirds of the mass at the wrist pin placed at radius $-r$ on the crank and recalculate the inertia force. Compare these results to those for the unbalanced crank.

†‡13-59 The footnote to the description of Figure 13-8c gives the maximum engine RPM for Nascar pushrod V-8 and Formula 1 V-12 and V-8 racing engines as 9600 and 19 000 RPM, respectively. Using the engine dimensions given in Figure 13-8, determine the peak accelerations in g's produced by engine speeds of 9600 and 19 000 RPM.

†‡13-60 Repeat Problem 13-38 with $\beta = 10°$ and $F_{g_{max}} = 1500$ lb using the engine dimensions and RPM given in Figure 13-8.

†‡13-61 Write a computer program or use an equation solver such as *Mathcad, Matlab*, or *TK-Solver* to calculate the approximately equivalent masses m_A and m_B for a crank-conrod-piston assembly. Test your program with the following data: $m_2 = 0.045$ blob, $m_3 = 0.025$ blob, $m_4 = 0.035$ blob, $r = 2$ in, $r_{G_2} = 0.8$ in, $l_a = 2.8$ in, and $l = 7$ in.

†‡13-62 Use the data from Problem 13-61 to calculate and plot the approximate inertia torque for one revolution of the crank with a crank speed of 1500 rpm.

†‡13-63 Using the gas force equation from Problem 13-38 with $\beta = 15°$, $F_{g_{max}} = 600$ lb, and the data from Problems 13-61 and 13-62, calculate and plot the total force on the wrist pin as the crank rotates from 0 to 180°.

†‡13-64 Use the data from Problem 13-62 to calculate and plot the main pin force for one revolution of the crank when it is a) exactly balanced and b) overbalanced with 60% of the mass m_B placed at $-r$ on the crank.

13.14 PROJECTS

These are loosely structured design problems intended for solution using program LINKAGES. All involve the design of a single-cylinder engine and differ only in the specific data for the engine. The general problem statement is:

Design a single-cylinder engine for a specified displacement and stroke cycle. Optimize the conrod/crank ratio and bore/stroke ratio to minimize shaking forces, shaking torque, and pin forces, also considering package size. Design your link shapes and calculate realistic dynamic parameters (mass, CG location, moment of inertia) for those links using the methods shown in Chapter 10 and Section 11.13. Dynamically model the links as described in this chapter. Balance or overbalance the linkage as needed to achieve these results. Overall smoothness of total torque is desired. Design and size a minimum weight flywheel by the method of Section 11.11 to smooth the total torque. Write an engineering report on your design.

P13-1 Two-stroke cycle with a displacement of 0.125 liter.

P13-2 Four-stroke cycle with a displacement of 0.125 liter.

P13-3 Two-stroke cycle with a displacement of 0.25 liter.

P13-4 Four-stroke cycle with a displacement of 0.25 liter.

P13-5 Two-stroke cycle with a displacement of 0.50 liter.

P13-6 Four-stroke cycle with a displacement of 0.50 liter.

P13-7 Two-stroke cycle with a displacement of 0.75 liter.

P13-8 Four-stroke cycle with a displacement of 0.75 liter.